



Decision Support for Future-Ready Digital Urban Governance: Evaluating Digital Public-Service Transformation Strategies in Indonesia

Muhammad Fahad Zaib¹, Shultonnyck Adha^{2*}, Ghazanfar Abbas³

¹ Department of Philosophy, Sociology, Education and Applied Psychology, University of Padua, 35139 Padua, Italy

² Management Department, Kusuma Negara Business School, 13770 Jakarta, Indonesia

³ Institute of Mathematics, University of the Punjab, 54590 Lahore, Pakistan

* Correspondence: Shultonnyck Adha (shultonnyck@gmail.com)

Received: 07-28-2026

Revised: 09-23-2026

Accepted: 10-03-2026

Citation: M. F. Zaib, S. Adha, and G. Abbas, "Decision support for future-ready digital urban governance: Evaluating digital public-service transformation strategies in Indonesia," *J. Urban Dev. Manag.*, vol. 5, no. 4, pp. 285–326, 2026. <https://doi.org/10.56578/judm050402>.



© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: Rapid urbanization and the expansion of artificial intelligence, digital platforms, connected infrastructure, and automated public services are reshaping how urban governments plan, coordinate, and deliver public services. In Indonesia, selecting an appropriate digital governance strategy remains challenging because expected service improvements must be considered alongside digital inclusion, public trust, algorithmic fairness, institutional capacity, environmental resilience, privacy risks, and implementation costs. This study investigates how alternative digital urban governance strategies can be evaluated within a transparent multi-criteria decision-support framework. Ten candidate strategies were evaluated against fifteen technical, social, institutional, environmental, and economic criteria using constructed role-based picture fuzzy assessments. Mutual information (MI) was applied to identify overlapping criteria and reduce redundancy, after which the Logarithmic Percentage Change-driven Objective Weighting (LOPCOW) method was used to derive objective criterion weights. Ranking Comparison (RANCOM) weights were then generated from the LOPCOW-based criterion ordering and combined with the objective weights before the alternatives were ranked using the Measurement of Alternatives and Ranking according to Compromise Solution (MARCOS) method. Ranking robustness was examined through parameter sensitivity analysis, 10,000-run Monte Carlo simulations, and comparisons with seven established multi-criteria decision-making (MCDM) methods, including the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), ViseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR), Multi-Objective Optimization on the Basis of Ratio Analysis (MOORA), Multi-Attributive Border Approximation Area Comparison (MABAC), Weighted Aggregated Sum Product Assessment (WASPAS), Evaluation based on Distance from Average Solution (EDAS), and Additive Ratio Assessment (ARAS). The results showed that mobile-first public services for remote and island communities ranked first, with a MARCOS appraisal score of 0.734412, followed closely by citizen participation and e-consultation at 0.723203. Privacy-preserving urban data exchange ranked third at 0.686119. The leading strategy remained stable across the tested parameter settings within the proposed MARCOS framework, although alternative weighting and ranking specifications produced greater variation in the identity of the first-ranked strategy. This result, together with the narrow gap between the two highest-ranked alternatives, indicates that local priorities, assessment inputs, and modelling choices can materially influence the final decision. The findings demonstrate that the proposed framework provides a traceable basis for comparing competing digital urban governance strategies while making the assumptions, trade-offs, and uncertainty underlying the ranking explicit. The approach offers urban authorities a structured decision-support tool for planning inclusive, accountable, resilient, and institutionally feasible public-service transformation.

Keywords: Digital urban governance; Public-service transformation; Decision support; Multi-criteria decision making; Picture fuzzy sets; MARCOS; Digital inclusion; Indonesia

1 Introduction

For many residents, a digital public-service platform is now the starting point for interacting with government. It may be used to obtain an official document, check the progress of an application, or report a local problem. The way

such an interaction is designed affects not only the time required to complete a service but also whether residents can complete it independently and obtain a useful outcome. Luna et al. [1] examined the public value perceived by citizens in digital service delivery, while Dechamps et al. [2] distinguished different ways in which governments conceptualize citizen-centred services. Together, these studies highlight an important issue for urban governance: the mere availability of a digital channel does not reveal how effectively residents can use it or what value they ultimately obtain from it. Indonesian experience illustrates the distinction between providing a digital access point and securing its continued and meaningful use. In their analysis of Jakarta Kini (JAKI), Al-Kautsar Maktub et al. [3] associated citizens' intention to use and actual use of the platform with government capacity, information and service quality, computer self-efficacy, trust, perceived risk, and related adoption factors. Varastika et al. [4] evaluated the M-Paspor application using the DeLone–McLean information systems success model. Although JAKI and M-Paspor serve different administrative purposes and their findings cannot be transferred directly from one service to another, both studies point to the same planning concern: public authorities must consider what residents can actually accomplish through a proposed digital service and how reliably the service supports that task. These considerations become particularly important when urban authorities must choose among a unified public-service portal, a mobile-first service, and more specialized digital governance platforms.

The intended users of public services do not begin from the same social, technological, or geographic position. A resident may own a mobile phone but lack reliable connectivity, digital confidence, language support, or the assistance required to complete an online transaction. Liu et al. [5]'s review of digital inclusion among vulnerable groups considered access together with skills, support mechanisms, and the outcomes of public-service use. Djabatmiko et al. [6] investigated how marginalized groups interact with digital public services and the barriers they experience. These findings provide a basis for identifying which groups are likely to benefit from a proposed digital service and which may remain disadvantaged. For many residents, for example, a mobile-first service may reduce the burden of travelling to a government office, whereas others may continue to depend on assistance or in-person contact with public employees. Such differences should therefore be incorporated into the evaluation of digital-service strategies rather than treating the existence of a mobile application as sufficient evidence of inclusion. Local governments also face institutional and resource constraints when introducing new digital solutions. Purnamasari et al. [7] examined the relationships among technological infrastructure, financial resources, digital innovation, public-service improvement, and government performance in Indonesia. Comparative research on digital governance in Association of Southeast Asian Nations (ASEAN) countries has likewise identified substantial variation in e-government development, together with continuing challenges related to internet access, digital literacy, and public trust [8]. Although these studies differ in their units of analysis, they converge on a decision faced by cities and other local authorities: the anticipated advantages of a digital system must be weighed against the financial, organizational, technical, and coordination requirements needed to implement and sustain it. A cross-agency platform, for instance, depends on interoperable systems and continuing institutional collaboration, whereas a more limited alternative may be easier to deploy but address only part of the underlying service problem. A meaningful evaluation therefore needs to make these trade-offs explicit.

The increasing use of artificial intelligence (AI) introduces an additional layer of concern into digital public-service planning. AI-enabled systems may assist public employees in classifying service requests, identifying patterns in administrative data, or predicting future service demand. At the same time, such systems may influence whether a resident is considered eligible for a service, how requests are prioritized, what information residents can access, and whether automated decisions can be reviewed or challenged. Agbabiaka et al. [9] examined the requirements for trustworthy AI-enabled automated decision-making in public organizations. De Almeida and dos Santos Júnior [10] investigated how public organizations govern the implementation and use of AI. Their findings support the need to assess an AI-enabled proposal not only in terms of technical capability but also in relation to an organization's capacity to regulate its use, explain automated processes, and govern the information on which those processes depend. Expected gains in efficiency must consequently be considered alongside equity, privacy, accountability, and the possibility of error, none of which is resolved simply through the adoption of AI. Implementation assessments should also reflect how services are experienced by residents. Catalá-Pérez et al. [11] used a citizen-led mystery-shopper exercise together with qualitative comparative analysis to examine digital-service encounters. Their approach draws attention to the sequence between locating a service and receiving a useful response. At the strategic level, Halim and Bounabat [12] proposed linking multidimensional digital-government assessment to subsequent action. The present study addresses a more specific decision within this broader planning problem by comparing alternative digital urban governance strategies in terms of service access, efficiency, inclusion, institutional feasibility, cost, resilience, and risk under explicitly stated case-study assessments.

No single measure can adequately represent all of these considerations. Some criteria may also overlap. Greater service accessibility, for example, may improve service efficiency, and evaluating both without examining their dependence may unintentionally give one underlying dimension excessive influence. Different role-based assessment profiles may use the same criteria while assigning different assessments to the available strategies. For this reason,

the evaluation procedure used in this study keeps several stages conceptually distinct. The constructed role-based assessments are first represented and aggregated using specified profile weights; dependence among evaluation criteria is then examined; the retained criteria are weighted; and the resulting information is used to establish the final strategy ranking. Separating these stages makes the assumptions underlying the ranking more transparent and allows the source of each component of the decision process to be traced. The procedure is intended to support comparison and planning rather than to predict what will necessarily occur after a strategy is implemented.

The study therefore pursues the following objectives:

1. to define candidate digital urban governance strategies and the corresponding benefit and cost criteria used to assess them in the Indonesian case;
2. to represent four constructed role-based linguistic assessment profiles as picture fuzzy numbers (PiFNs) and aggregate them using competence-based Softmax weights;
3. to identify overlapping criteria through pairwise mutual information (MI) and maintain a consistent index mapping for the retained criteria;
4. to calculate Logarithmic Percentage Change-driven Objective Weighting (LOPCOW) objective weights and derive Ranking Comparison (RANCOM) ordinal weights from the LOPCOW ordering, without treating the latter as an independent source of subjective constructed assessment;
5. to combine the criterion weights and rank the candidate strategies using the Measurement of Alternatives and Ranking according to Compromise Solution (MARCOS) method while reporting the intermediate quantities required to trace the calculation; and
6. to examine the dependence of the resulting ranking on the assessments, model parameters, and ranking assumptions before drawing implications for policy use.

1.1 Foundations of Fuzzy Set Theory and Multi-Criteria Decision-Making

Many policy and management decisions begin with assessments that are inherently imprecise and cannot be represented adequately by a sharp boundary between two states. A digital public service, for example, may be readily accessible to some residents but only partly accessible to others. The distinction may depend on distance from service facilities, communication infrastructure, disability, language, digital skills, and the particular task that must be completed. Zadeh's fuzzy-set formulation introduced a means of representing gradual degrees of membership through values between 0 and 1 [13]. This allows qualitative descriptions such as low, moderate, and high to be represented without forcing every assessment into a binary classification. Fuzzy multi-criteria decision-making (MCDM) has since developed into a broad field concerned with the representation of imprecise assessments and with the difficulties of combining uncertain judgments, criterion weights, and ranking rules [14].

A conventional membership value records the degree of support for an assessment but does not independently represent opposition to it. Atanassov's intuitionistic fuzzy set addressed this restriction by assigning both a membership degree and a non-membership degree, subject to the condition that their sum does not exceed one [15]. The remaining proportion is generally interpreted as hesitation. This distinction is useful when a decision-maker identifies evidence both in favour of and against a proposal, because uncertainty does not have to be inferred solely from a low membership value. Recent applications of intuitionistic fuzzy information include disaster preparedness in clinical laboratories and feature selection under interval-valued assessments [16, 17]. Other studies have applied related approaches to operating conditions in aluminium production, sustainable airport development, and factors influencing the adoption of decentralized renewable energy [18–20]. Although these applications differ substantially in context, each involves multiple considerations for which a single crisp value may conceal important features of the underlying judgment.

Intuitionistic fuzzy sets still leave some evaluative positions implicit. An evaluator may, for example, adopt a neutral position rather than simply hesitate between support and opposition. Cuong's picture fuzzy sets explicitly represent neutrality, together with refusal. A picture fuzzy assessment contains positive, neutral, and negative membership degrees whose total cannot exceed one; the unassigned remainder represents the refusal degree [21]. Neutrality and refusal have different interpretations. Neutrality indicates an explicitly intermediate position, whereas refusal indicates that part of the assessment remains uncommitted. Recent picture fuzzy studies have examined how multiple forms of evaluation can be represented and aggregated when several responses are involved [22, 23]. Their interpretation, however, still depends on the question being asked and on the linguistic scale used to obtain the assessment.

MCDM provides the second component required for the present framework. It compares alternative courses of action using criteria that may favour different choices, determines how much importance should be attached to those criteria, evaluates the performance of alternatives, and establishes an overall ranking. Fuzziness can enter an MCDM model at four distinct stages. First, it may be used when alternatives or criteria are evaluated. Second, it may be introduced during criterion weighting. Third, it may be used when individual assessments are aggregated. Fourth, it may also affect the final procedure used to establish the preference ordering. These are separate modelling decisions and should not be treated as a single methodological choice. A recent picture fuzzy distance-based method,

for example, calculated relative closeness coefficients before applying a modified compromise-ranking procedure [24]. Other studies have addressed interaction among assessments and criteria through picture fuzzy aggregation combined with Dempster–Shafer theory [22], while picture hesitant fuzzy soft sets have been used to represent additional variability in assessments for sustainable solar-energy management [25]. Such extensions illustrate the range of modelling possibilities, but greater mathematical complexity does not in itself make a representation more appropriate for a particular decision problem.

The diversity of existing applications helps to locate the present study methodologically. Picture fuzzy assessments have been used to evaluate the quality of online medical services from patient comments [23], to analyse changes in public perceptions of transportation infrastructure [26], and to prioritize risks in construction and demolition waste management [27]. Related research has combined picture fuzzy assessments with Evaluation based on Distance from Average Solution (EDAS) for decision-making in healthcare and has used picture fuzzy information to select locations for disaster-response logistics hubs [28, 29]. Picture fuzzy concepts have also been extended to emergency-management decisions through complex Fermatean picture fuzzy models [30]. The substantive problems and mathematical extensions in these studies differ considerably, so their rankings cannot be transferred directly to a digital urban governance problem. Their broader relevance lies instead in showing how uncertain judgments can be linked to an explicit weighting and ranking procedure.

In digital urban governance, the evaluation must accommodate issues such as service accessibility, administrative feasibility, privacy, institutional readiness, environmental resilience, and implementation cost. Picture fuzzy scales make it possible to preserve positive, neutral, and negative components of a judgment during group aggregation. Once aggregated, however, the information must still be transformed in a manner compatible with the subsequent criterion-screening, weighting, and ranking procedures. These stages address different questions: whether criteria contain overlapping information, how much importance each retained criterion receives, and which strategy performs best under the resulting decision model. Keeping these stages separate makes it easier to trace how the final ordering was obtained. Accordingly, the present study uses picture fuzzy assessments to represent the initial evaluations and then applies a transparent MCDM sequence to screen criteria, derive criterion weights, and rank alternative digital urban governance strategies. The resulting rankings remain conditional on the stated assessments and modelling assumptions.

1.2 Multi-Criteria Decision-Making Methods in Recent Decision-Support Studies

Selecting a digital public-service strategy involves more than identifying the alternative that appears fastest or most technologically advanced. An automated service may increase processing speed or extend service coverage while simultaneously creating new operating costs, accessibility concerns, institutional demands, and risks for residents. Multi-criteria decision-making is therefore useful because it requires the alternatives, evaluation criteria, assessment information, criterion importance, and ranking procedure to be specified explicitly. Recent urban and infrastructure studies demonstrate the range of approaches available. A study of sustainable smart-city determinants used interpretive structural modelling together with the Analytic Hierarchy Process (AHP) to organize and weight indicators [31]. Research on sustainable urban mobility linked traffic simulation with multi-criteria assessment [32], while another urban-planning study employed Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE) within a visual decision environment [33]. Elsewhere, AHP, Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), VIseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR), Complex Proportional Assessment (COPRAS), and a weighted-sum model were compared on the same passive-design decision problem [34]. Collectively, these studies demonstrate that an MCDM method should be selected in relation to the decision it is intended to support rather than treated as an interchangeable computational procedure.

Table 1 summarizes twelve recent decision-support applications published between 2024 and 2026. They cover criterion structuring, subjective and objective weighting, distance-based ranking, compromise solutions, outranking, and proportional assessment. The substantive findings of these studies cannot be transferred directly to Indonesian digital public services because each application uses its own alternatives, assessment evidence, and definition of a desirable outcome. They nevertheless provide useful methodological reference points for designing and validating an urban governance assessment.

The Indonesian case study adopts a more specific computational sequence. MI is used to screen criteria, LOPCOW derives objective weights from the assessed matrix, and MARCOS ranks the candidate strategies. The RANCOM comparison inputs are constructed from the ordering produced by LOPCOW. The resulting RANCOM values therefore constitute a transformation of the objective-weight ordering rather than an independent expression of expert preference. This dependence must remain explicit when the combined weights are interpreted. Comparisons with alternative ranking procedures can subsequently indicate whether the ordering is sensitive to the computational rule applied to the same case-study matrix; they cannot establish how a strategy will perform after implementation by a municipality.

Table 1. Recent MCDM studies and the methods used in their decision models

Study	Decision Problem	Assessment Information	Structuring or Weighting	Evaluation Method	Use for an Urban Governance Study
Smart-city determinants [31]	Sustainable smart-city indicators	Constructed assessments and indicators	Interpretive structural modelling and AHP	Indicator priorities	Shows how relationships among indicators can be examined before assigning priorities.
Urban mobility DSS [32]	Sustainable mobility policies	Traffic simulation and policy criteria	Multi-criteria decision model	Policy assessment	Connects a decision model to simulated service outcomes.
Urban planning model [33]	Planning options in a visual urban environment	Spatial and other urban information	Explicit criteria and preferences	PROMETHEE	Illustrates pairwise outranking when planners need to inspect preferences.
Passive-design comparison [34]	Building-design strategies	Performance on design criteria	AHP and alternative weight settings	COPRAS, TOPSIS, VIKOR, and weighted sum	Demonstrates that a recommendation can be checked across several ranking rules.
Municipal waste scenarios [35]	Solid-waste management options	Fuzzy assessments and scenario criteria	AHP	TOPSIS and PROMETHEE II	Combines uncertain judgments with distance-based and outranking comparisons.
Circular smart cities [36]	Circular-economy priorities	Expert assessments of urban criteria	Neuro decision-making model	VIKOR, checked against TOPSIS	Provides an example of compromise ranking and a second-method check.
Urban traffic resilience [37]	Importance of road intersections	Network and traffic characteristics	Entropy, CRITIC, AHP, BWM, and other weights	TOPSIS, VIKOR, EDAS, MARCOS, ELECTRE III, and others	Shows how both weighting and ranking choices can affect a network assessment.
Charging-station locations [38]	Sites for electric-vehicle charging	Technical, environmental, and economic criteria	DEMATEL and ANP	COPRAS	Models interactions among criteria before proportional ranking.
Blockchain adoption [39]	Factors affecting carbon-trading adoption	Expert assessments of factor influence	DEMATEL-based ANP	Factor priorities	Useful when institutional and technical criteria influence one another.
Retail strategy selection [40]	Marketing strategies	Fuzzy expert assessments	Fuzzy DEMATEL	COPRAS	Separates analysis of criterion influence from comparison of alternatives.
Public-transport selection [41]	Citizen-informed transport options	Citizen-derived criterion judgments	AHP	SAW, TOPSIS, VIKOR, and PROMETHEE II	Illustrates how public preferences can enter weighting and how rankings can be compared.
Wind-farm siting [42]	Candidate wind-farm sites	Site performance and criterion judgments	AHP, BWM, entropy, CRITIC, and other weights	TOPSIS, VIKOR, ELECTRE III, and PROMETHEE II	Tests combinations of weighting and ranking methods for a location decision.

Note: MCDM = multi-criteria decision-making; DSS = decision support system; AHP = Analytic Hierarchy Process; PROMETHEE = Preference Ranking Organization Method for Enrichment Evaluation; COPRAS = Complex Proportional Assessment; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; VIKOR = VIsse Kriterijumska Optimizacija I Kompromisno Resenje; CRITIC = Criteria Importance Through Intercriteria Correlation; BWM = Best-Worst Method; EDAS = Evaluation based on Distance from Average Solution; MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; ELECTRE = ELimination Et Choix Traduisant la REALité; DEMATEL = Decision-Making Trial and Evaluation Laboratory; ANP = Analytic Network Process; SAW = Simple Additive Weighting.

1.3 Digital Urban Governance and Artificial Intelligence-Enabled Public Services in Indonesia

Research on digital government in Indonesia indicates that the challenge extends well beyond the introduction of new applications. Pribadi et al. [43]'s study of digital-government service implementation identified relationships involving employee skills, leadership, regulation, infrastructure, transparency, and accountability. Research on digital innovation adoption has likewise considered technological infrastructure and financial resources together with public-service and government-performance outcomes [7]. Although these studies differ in their samples and measures, both place institutional capacity at the centre of digital-service implementation. A strategy may therefore perform strongly in terms of technological capability while remaining difficult for a local authority to finance, coordinate, or operate.

Citizens' experience provides another dimension of service performance. Research on Jakarta Kini (JAKI) examined the adoption of a municipal application and identified government capacity, information and service quality, trust, perceived risk, and users' ability to work with computers among the factors associated with intention and use [3]. A separate Indonesian e-government study applied the DeLone and McLean model and found that service quality was related to user satisfaction [44]. Because the studies examine different services, their findings should not be interpreted as a single estimate of public acceptance. Taken together, however, they show why the assessment of urban digital strategies needs to consider whether residents can use and trust a service in addition to whether the service is technically available.

The movement toward AI-enabled administration raises further questions about how digital systems are integrated into established public-sector processes. Existing research has examined the evidence base for AI development in Indonesia's public sector, although the available findings remain distributed across different applications and administrative settings [45]. A study of AI adoption in Indonesian public-sector organizations considered service requests through the organization's business architecture [46]. Its emphasis is therefore on the data, processes, and organizational roles required to integrate AI into public administration rather than on an isolated technological tool. Research in Makassar has also examined policy networks surrounding AI-enabled governance, including the roles of local government, technology providers, academia, and citizens [47]. These studies use different methodological approaches: literature reviews map the existing evidence base, architecture studies develop organizational frameworks, and governance studies examine relationships among participating actors. None, however, provides a direct basis for determining which of several competing city-level digital-service investments should receive priority.

More recent work has focused particularly on readiness and regulatory oversight. Drawing on surveys of local-government employees in Indonesia, Widuri and Jarkasih [48] reported a relationship between organizational AI readiness and digital public-service innovation. Research on AI-regulatory oversight in Indonesia has examined administrative authority, algorithmic auditing, and accountability [49]. Studies of digital government following administrative reform have also considered the responsibilities of state institutions in implementing public services [50]. Organizational readiness, regulatory authority, and accountability cannot be represented adequately by a single measure of technological performance. They relate to different stages of the decisions that municipalities must make before and after implementation. Existing research therefore provides valuable evidence on technology adoption, user experience, institutional conditions, and governance arrangements, but it offers comparatively little guidance on how an Indonesian municipal authority might compare several proposed service strategies when the relevant considerations conflict.

1.4 Motivation of the Study

An Indonesian municipality facing limited resources may need to decide which form of digital public-service development should receive priority: a shared service portal, mobile-first services for remote areas, a citizen-engagement platform, inter-agency data exchange, or another digitally enabled governance strategy. Each alternative has different implementation costs, technical requirements, institutional demands, and implications for public trust. AI may support some components of these services, but its effectiveness depends on the availability and quality of historical data, the degree of decision authority assigned to automated systems, and the procedures established to identify, review, and correct errors. A municipal authority therefore needs to evaluate these alternatives as service and governance strategies rather than as technology-procurement options alone.

Prior research has identified many factors that affect the ability of municipalities and other public institutions to implement digital government successfully. Organizational capacity, infrastructure, transparency, and accountability have been identified as important conditions for digital-government performance [43]. Other studies have examined the implementation and use of citizen-facing digital applications through factors such as user adoption, trust in government, perceived service quality, and overall satisfaction [3, 44]. More recent work has focused on whether public institutions are prepared to use AI and on the responsibilities required to govern its application [48, 49]. These studies describe important dimensions of digital transformation, but they do not establish a common analytical basis for comparing competing service strategies in which wider service reach may need to be weighed against privacy protection, institutional feasibility, or implementation burden.

Such comparisons involve different forms of evidence. An estimated implementation cost may be represented numerically, whereas institutional readiness may depend on judgments concerning staffing, inter-agency coordination, legal authority, and organizational capability. Residents may value easier access to services while simultaneously expressing concern about how government uses their personal data. Indonesia's geographic and infrastructural diversity further complicates such decisions. A service strategy based on smartphones and continuous internet connectivity, for example, may impose very different burdens in a metropolitan area and in a remote or island community. Treating these differences as secondary considerations after a strategy has already been selected would leave a substantial part of the decision unexplained.

A second problem is the possibility of double-counting related information. Service accessibility and digital inclusion, for instance, may partly reflect the same underlying evidence. If closely related criteria are rewarded independently without first examining their dependence, one underlying concern may receive disproportionate influence. Conversely, a criterion that varies little across the alternatives may contribute little discriminatory information even if it remains substantively important. Criterion screening and weighting make these issues visible, but the resulting weights must still be interpreted carefully. In the present study, the RANCOM comparisons were generated from the LOPCOW ordering and therefore do not constitute independently elicited expert preferences.

Against this background, the study develops a decision-support procedure for evaluating AI-enabled and digitally mediated public-service strategies in the Indonesian urban-governance context. Picture fuzzy assessments are used to preserve positive, neutral, and negative components of the initial evaluations. MI is then applied to examine redundancy among criteria, LOPCOW is used to derive objective criterion weights, RANCOM transforms the resulting ordering into comparison-based weights, and MARCOS establishes the final ranking of the candidate strategies. The purpose is not to claim that one digital governance strategy is universally preferable for Indonesia, but to show how a municipal authority can document the assumptions supporting a recommendation, identify closely competing alternatives, and examine the sensitivity of the resulting order. The assessments used in the present case are constructed for methodological demonstration. Before the framework is used to support an actual investment decision, the illustrative inputs would need to be replaced with locally collected information from public services, residents, responsible agencies, and other relevant stakeholders.

1.5 Research Gap

Two related gaps motivate this study. The first is methodological and concerns how an MCDM model should handle uncertain assessments, overlapping criteria, and the provenance of criterion weights. The second concerns the decision context itself. Research on Indonesian digital government provides substantial evidence on digital-service adoption, institutional capacity, user experience, and AI governance, yet those findings are rarely translated into directly comparable assessments of competing urban public-service strategies. Tables 2 and 3 summarize these two gaps and show how the present case study uses the available evidence without extending it beyond what the cited studies can support. The proposed procedure addresses a specific comparison problem; it does not eliminate the limitations inherent in the selected methods or in the constructed case-study inputs.

The methodological contribution lies in the traceable sequence applied to this particular comparison. Picture fuzzy assessments retain the three stated forms of judgment. MI is used to examine relationships among criteria before weighting, LOPCOW derives objective weights from the assessed matrix, and MARCOS produces the final strategy ordering. RANCOM is retained with an explicit qualification: because its comparisons follow the LOPCOW ordering, it cannot be interpreted as an independent expert-weighting stage. Making this dependence explicit prevents the transformed RANCOM vector from being presented as a second independent source of evidence.

The case-study gap lies between knowledge about individual digital services and the practical choice among several possible investments. Research on JAKI identifies concerns associated with use and trust, while studies of institutions, resources, and AI adoption identify conditions required for digital services to operate. None of these findings provides a ready-made numerical score for a mobile-first service, a shared public-service portal, a citizen-participation platform, or a data-exchange strategy. The present study therefore places the proposed alternatives within a common decision matrix and specifies how their assessed performance is translated into a ranking.

The resulting ordering must be interpreted within this boundary. The case-study judgments were constructed to examine the decision procedure and are not observations of implemented strategies across Indonesian municipalities. The study addresses the comparison gap by showing which forms of evidence would need to be brought together, how overlapping criteria and criterion weights can be handled, and how closely the competing strategies are positioned under the stated inputs. Application to a real municipal investment decision would require the same framework to be populated with locally collected assessments and observed service information.

1.6 Structure of the Paper

The remainder of the paper is organized as follows. Section 2 introduces the picture fuzzy set concepts used to represent the assessments. Section 3 presents the decision-support methodology, including group aggregation,

criterion screening, weighting, and ranking. Section 4 describes the Indonesian digital urban governance case study and reports the corresponding assessment inputs and analytical results. Section 5 discusses the findings, their implications for urban governance, and the principal limitations of the analysis. Section 6 concludes the study and identifies directions for future research.

Table 2. Methodological gaps and the response in this study

Method or Study	Main Contribution	Limitation for the Present Decision	Why the Limitation Matters	Response in This Study	Remaining Limitation
AHP-based smart-city assessment [31]	Structures indicators and assigns priorities through comparisons	Its indicator priorities do not provide assessments of the governance strategies considered here	A criterion framework alone cannot determine which proposed service strategy ranks first	Defines the strategies and assesses each against a common set of criteria	Results still depend on the selected strategies and criteria
Fuzzy AHP, TOPSIS, and PROMETHEE in municipal waste planning [35]	Represents uncertain judgments and compares management scenarios	Its assessments and weights concern a different public-service problem	A published ranking cannot be transferred to digital governance without new inputs	Uses picture fuzzy assessments to retain positive, neutral, and negative judgments about the strategies	The linguistic assessments are constructed for the case study
Multi-method comparison of passive-design options [34]	Examines how several ranking rules behave on a common decision problem	Agreement among methods concerns the matrix and weights used in that study	Agreement among methods does not establish that the inputs describe actual service outcomes	States the assessment inputs and calculates a MARCOS ranking for the Indonesian case	Comparisons with other methods must use the same case inputs and be reported separately
Alternative weighting methods in urban traffic assessment [37]	Compares weights derived from data and from expressed judgments	Different sources of weights cannot be treated as independent unless they are obtained independently	Combining dependent weights may give a misleading impression of expert involvement	Reports the LOPCOW weights and identifies the RANCOM comparison matrix as being derived from their ordering	The RANCOM vector is not independent subjective evidence
Citizen-weighted, multi-method transport assessment [41]	Uses citizen input for weights and compares several ranking methods	Citizen-derived transport weights cannot be substituted for judgments concerning the governance criteria used here	Preferences depend on both the participants and the decision presented to them	Makes the origin of every weight explicit in the case-study procedure	No citizen-preference data were collected for the present assessment

Note: AHP = Analytic Hierarchy Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; PROMETHEE = Preference Ranking Organization Method for Enrichment Evaluation; MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; RANCOM = Ranking Comparison.

2 Preliminaries

Picture fuzzy sets provide a flexible representation for assessments involving simultaneous support, neutrality, and opposition. This feature is particularly relevant to digital urban governance decisions, where a proposed public-service strategy may be viewed favourably in one respect, neutrally in another, and critically in relation to a separate concern. These three components are subject to a joint constraint, ensuring that each assessment constitutes a valid PiFN.

Table 3. Indonesian case-study evidence and the decision gap addressed

Study	Evidence Examined	Findings in the Original Setting	Unresolved Question for This Study	Use in the Present Study	Limits of Transferability
Pribadi et al. [43]	Organizational and governance conditions for Indonesian digital services	Examines implementation alongside skills, infrastructure, transparency, and accountability	How should these conditions enter a comparison of proposed strategies?	Represents institutional feasibility and accountability in the assessment criteria	Reported relationships are not converted directly into numerical strategy scores
Purnamasari et al. [7]	Infrastructure, financial resources, digital innovation, and public-service performance	Studies resource conditions associated with digital-service outcomes	Can an option with strong expected service reach be supported with available resources?	Includes implementation demands and cost alongside service-related benefits	The study's measured effects are not reused as criterion weights
Jakarta Kini study [3]	Citizen acceptance and use of the JAKI application	Examines trust, perceived risk, information quality, and user capability	How should likely user experience be considered when broader service strategies are compared?	Assesses access, trust, and service quality as separate concerns	JAKI adoption results cannot predict adoption of every proposed digital service
AI architecture study [46]	AI adoption in Indonesian public-sector processes	Examines the fit among AI tools, service requests, and organizational architecture	Which proposals depend on difficult data or process integration?	Considers technical integration and administrative readiness	An adoption framework does not establish local implementation capacity
Makassar AI governance study [47]	Actors and coordination in AI-enabled public-service governance	Describes the roles of government and other participants in Makassar	How should responsibility across agencies affect the choice of strategy?	Includes coordination and governance among the assessed concerns	A network observed in Makassar is not assumed to exist in other cities

Note: JAKI = Jakarta Kini; AI = artificial intelligence.

Definition 2.1 (Picture fuzzy set) [21]. Let X be a nonempty universe. A picture fuzzy set P on X is

$$P = \{\langle x, \mu_P(x), \eta_P(x), \nu_P(x) \rangle : x \in X\} \quad (1)$$

where, $\mu_P(x)$, $\eta_P(x)$, and $\nu_P(x)$ denote the positive, neutral, and negative membership degrees, respectively. For every $x \in X$,

$$0 \leq \mu_P(x), \eta_P(x), \nu_P(x) \leq 1, \quad \mu_P(x) + \eta_P(x) + \nu_P(x) \leq 1. \quad (2)$$

The unassigned proportion, referred to as the refusal degree $\rho_P(x)$, is defined as

$$\rho_P(x) = 1 - \mu_P(x) - \eta_P(x) - \nu_P(x). \quad (3)$$

This quantity is referred to as the refusal degree. A triple $a = (\mu_a, \eta_a, \nu_a)$ satisfying Eq. (2) is termed a PiFN.

Definition 2.2 (Score and accuracy functions) [51]. For a PiFN $a = (\mu_a, \eta_a, \nu_a)$, its score function $s(a)$ is defined as

$$s(a) = \mu_a - \eta_a - \nu_a, \quad -1 \leq s(a) \leq 1, \quad (4)$$

and its accuracy function $h(a)$ is defined as

$$h(a) = \mu_a + \eta_a + \nu_a, \quad 0 \leq h(a) \leq 1. \quad (5)$$

The score function assigns a positive contribution to positive membership and subtracts the neutral and negative components. The accuracy function measures the total assigned degree and is used only when two PiFNs have identical score values.

For two PiFNs a and b , the comparison adopted in this study is lexicographic:

$$a \succ b \iff s(a) > s(b) \quad \text{or} \quad [s(a) = s(b) \text{ and } h(a) > h(b)] \quad (6)$$

where, $\succ$ and $\prec$ denote strict preference and reverse strict preference, respectively. Similarly, $a \prec b$ if $s(a) < s(b)$, or if $s(a) = s(b)$ and $h(a) < h(b)$. If both the score and accuracy values are identical, the two PiFNs are treated as equivalent under this comparison rule. The accuracy value therefore does not replace the score function; it serves only as a tie-breaking criterion.

3 Methodology

The selection of a future digital urban governance strategy is formulated as a group multi-criteria decision problem. Ten candidate strategies are evaluated against fifteen criteria representing technical, social, institutional, environmental, and economic considerations. Four role-based assessment profiles are constructed to represent distinct professional perspectives relevant to the decision problem: urban governance and public administration, information technology and cybersecurity, public policy and citizen behaviour, and urban infrastructure and environmental planning. These profiles are methodological constructs used to demonstrate the proposed decision-support procedure; they do not represent evaluations collected from recruited experts or identifiable human participants. PiFNs are used to represent the assessments because they allow positive, neutral, and negative judgments to be retained simultaneously while also accommodating an uncommitted component.

The proposed decision-support procedure consists of four connected stages. First, individual linguistic assessments are transformed into PiFNs and aggregated into a group decision matrix using competence-based assessment-profile weights. Second, MI is used to identify strongly overlapping criteria and reduce redundancy before weighting. Third, LOPCOW derives objective criterion weights from the reduced decision matrix, after which RANCOM converts the LOPCOW ordering into comparison-based weights. These two related weight vectors are then combined. Finally, MARCOS ranks the candidate digital governance strategies relative to ideal and anti-ideal reference profiles.

Let $U = \{G_1, G_2, \dots, G_m\}$ denote the set of candidate digital governance strategies, where G_i represents the i th strategy ($i = 1, 2, \dots, m$) and $m = 10$. Let $D = \{D_1, D_2, \dots, D_n\}$ denote the set of original evaluation criteria, where D_j represents the j th criterion ($j = 1, 2, \dots, n$) and $n = 15$. The role-based assessment profiles are denoted by $E = \{E_1, E_2, \dots, E_q\}$, where E_h represents the h th assessment profile ($h = 1, 2, \dots, q$) and $q = 4$.

The first eight criteria are benefit criteria, whereas the remaining seven represent costs, implementation burdens, or risks. The linguistic scale is expressed in favourable terms for all criteria. Consequently, for a cost criterion, a high linguistic rating denotes a comparatively low or manageable burden. Before LOPCOW weighting and MARCOS ranking are performed, the original cost direction is restored so that the mathematical treatment of benefit and cost criteria remains consistent.

3.1 Group Assessment and Aggregation of Digital Governance Strategies

Step 1: Construction of the linguistic decision matrices

For the methodological case study, each role-based assessment profile assigns a linguistic evaluation to every candidate strategy under every original criterion. The resulting linguistic assessment matrix associated with profile E_h is

$$L^{(h)} = \left[\ell_{ij}^{(h)} \right]_{m \times n}, \quad h = 1, 2, \dots, q \quad (7)$$

where, $\ell_{ij}^{(h)}$ denotes the constructed linguistic evaluation assigned under profile E_h to strategy G_i with respect to criterion D_j . Eq. (7) therefore generates four separate illustrative assessment matrices rather than a single pooled matrix. Keeping these matrices separate at this stage preserves the differences intentionally represented by the four professional perspectives until group aggregation is performed. The linguistic terms and their corresponding picture fuzzy representations are listed in Table 4. A PiFN is written as $\langle a, b, c \rangle$, where a , b , and c denote the positive, neutral, and negative membership degrees, respectively.

Step 2: Transformation into picture fuzzy assessments

Each linguistic entry $\ell_{ij}^{(h)}$ is replaced by its corresponding PiFN from Table 4:

$$Z_{ij}^{(h)} = \langle a_{ij}^{(h)}, b_{ij}^{(h)}, c_{ij}^{(h)} \rangle \quad (8)$$

where, $Z_{ij}^{(h)}$ denotes the picture fuzzy evaluation assigned by profile E_h to strategy G_i under criterion D_j , and $a_{ij}^{(h)}$, $b_{ij}^{(h)}$, and $c_{ij}^{(h)}$ represent its positive, neutral, and negative membership degrees, respectively.

Table 4. Linguistic scale used to assess digital urban governance strategies

Evaluation Level	Code	PiFN
Fully suitable for future urban governance	FSG	(0.90, 0.03, 0.03)
Highly suitable for public-service transformation	HSP	(0.82, 0.06, 0.07)
Suitable with minor implementation concerns	SMC	(0.74, 0.09, 0.12)
Moderately suitable for urban implementation	MSI	(0.65, 0.13, 0.17)
Acceptable under existing conditions	AEC	(0.55, 0.18, 0.22)
Limited suitability for public-service delivery	LSP	(0.45, 0.22, 0.28)
Poorly suited to current institutional capacity	PSC	(0.34, 0.27, 0.34)
Unsuitable for urban governance implementation	UGI	(0.22, 0.30, 0.43)

Note: PiFN = picture fuzzy number.

The three membership components satisfy

$$0 \leq a_{ij}^{(h)} + b_{ij}^{(h)} + c_{ij}^{(h)} \leq 1. \quad (9)$$

The unassigned component, referred to as the refusal degree $d_{ij}^{(h)}$, is calculated as

$$d_{ij}^{(h)} = 1 - a_{ij}^{(h)} - b_{ij}^{(h)} - c_{ij}^{(h)}. \quad (10)$$

Eqs. (9) and (10) are checked for every entry before the individual decision matrices are aggregated.

Step 3: Construction of profile-specific competence scores

The four role-based perspectives used in the methodological case study are summarized in Table 5. These profiles represent professional viewpoints relevant to the decision model and are not descriptions of recruited human participants. Four competence dimensions are used to differentiate the profiles computationally: professional experience, knowledge of the decision problem, technical competence, and familiarity with the Indonesian context. The resulting competence values are constructed methodological inputs used to demonstrate the weighting procedure rather than empirically observed characteristics of individual experts.

$$R_{hr} = \langle u_{hr}, v_{hr}, z_{hr} \rangle$$

denotes the picture fuzzy competence specification of assessment profile E_h under competence dimension r , where u_{hr} , v_{hr} , and z_{hr} represent the positive, neutral, and negative membership degrees, respectively. Its scalar score is

$$e_{hr} = u_{hr} - v_{hr} - z_{hr}. \quad (11)$$

Table 5. Role-based professional perspectives represented in the constructed assessment framework

Role	Responsibility in the Assessment	Relevant Knowledge
Urban governance and public administration specialist	Reviews administrative feasibility, regulatory requirements, institutional coordination, and public-service delivery	Public administration, regulatory analysis, urban governance, and public-sector digital programmes
Information technology and cybersecurity specialist	Examines infrastructure requirements, system integration, data protection, and exposure to technical failure or cyberattack	Information systems, cybersecurity, artificial intelligence, digital identity, and database integration
Public policy and citizen-behaviour specialist	Assesses trust, acceptance, inclusion, fairness, participation, and the social effects of automated public services	Public policy, behavioural analysis, technology acceptance, citizen engagement, and ethical governance
Urban infrastructure and environmental planning specialist	Reviews infrastructure compatibility, implementation demands, environmental effects, disaster preparedness, and climate response	Urban planning, smart infrastructure, environmental management, and climate-responsive development

Eq. (11) transforms each picture fuzzy competence assessment into a scalar before the four competence dimensions are combined. If ρ_r denotes the importance assigned to competence dimension r , the overall competence score of assessment profile E_h is

$$e_h = \sum_{r=1}^4 \rho_r e_{hr}, \quad \rho_r \geq 0, \quad \sum_{r=1}^4 \rho_r = 1. \quad (12)$$

Equal competence-dimension weights, $\rho_r = 0.25$, are used when no empirical basis supports a different allocation.

Step 4: Calculation of assessment-profile weights

The competence scores obtained from Eq. (12) are converted into strictly positive assessment-profile weights w_h using a Softmax transformation:

$$w_h = \frac{\exp(e_h/\tau)}{\sum_{t=1}^q \exp(e_t/\tau)}, \quad h = 1, 2, \dots, q \quad (13)$$

where, $\tau > 0$ is a concentration parameter. Smaller values of τ assign relatively greater numerical influence to profiles assigned higher competence scores, whereas larger values move the distribution closer to equal weighting. The value of τ is reported together with the computational results. The resulting weights satisfy

$$w_h > 0, \quad \sum_{h=1}^q w_h = 1.$$

Step 5: Aggregation of picture fuzzy evaluations

For each strategy–criterion pair (i, j) , the four picture fuzzy evaluations are combined using a Dombi weighted averaging operator. The aggregated picture fuzzy evaluation of strategy G_i under criterion D_j is denoted by

$$\bar{Z}_{ij} = \langle \bar{a}_{ij}, \bar{b}_{ij}, \bar{c}_{ij} \rangle.$$

With Dombi parameter $\kappa > 0$, the positive membership component $\bar{a}_{ij}$ is calculated as

$$\bar{a}_{ij} = 1 - \frac{1}{1 + \left[\sum_{h=1}^q w_h \left(\frac{a_{ij}^{(h)}}{1 - a_{ij}^{(h)}} \right)^\kappa \right]^{1/\kappa}}, \quad (14)$$

the neutral membership component $\bar{b}_{ij}$ as

$$\bar{b}_{ij} = \frac{1}{1 + \left[\sum_{h=1}^q w_h \left(\frac{1 - b_{ij}^{(h)}}{b_{ij}^{(h)}} \right)^\kappa \right]^{1/\kappa}}, \quad (15)$$

and the negative membership component $\bar{c}_{ij}$ as

$$\bar{c}_{ij} = 1 - \frac{1}{1 + \left[\sum_{h=1}^q w_h \left(\frac{c_{ij}^{(h)}}{1 - c_{ij}^{(h)}} \right)^\kappa \right]^{1/\kappa}}. \quad (16)$$

Eqs. (14)–(16) are applied to all mn strategy–criterion positions to obtain the aggregated picture fuzzy group matrix $\bar{Z}$:

$$\bar{Z} = [\bar{Z}_{ij}]_{m \times n}. \quad (17)$$

Step 6: Transformation of the aggregated matrix into crisp scores

After aggregation, each group PiFN $\bar{Z}_{ij}$ is converted into a scalar score x_{ij} :

$$x_{ij} = \bar{a}_{ij} - \bar{b}_{ij} - \bar{c}_{ij}. \quad (18)$$

The resulting crisp decision matrix X is

$$X = [x_{ij}]_{m \times n}. \quad (19)$$

Eq. (18) is applied only after the individual evaluations have been aggregated. Applying the score function to the four assessment-profile matrices before aggregation would discard part of the picture fuzzy information before a group assessment is formed. The minimum, maximum, mean, and standard deviation of each column of Eq. (19) are examined to identify constant or malformed criterion values before subsequent processing.

Step 7: Estimation of inter-criterion dependence

Column j of X is denoted by

$$\mathbf{x}_j = (x_{1j}, x_{2j}, \dots, x_{mj})^T.$$

For every criterion pair (j, k) , the MI between criteria D_j and D_k , denoted by I_{jk} , is defined as

$$I_{jk} = \iint p_{jk}(s, t) \log \left[\frac{p_{jk}(s, t)}{p_j(s)p_k(t)} \right] ds dt \quad (20)$$

where, s and t denote the values of criteria D_j and D_k , respectively; $p_{jk}(s, t)$ is their joint probability density, while $p_j(s)$ and $p_k(t)$ are the corresponding marginal densities.

Eq. (20) is estimated using a k -nearest-neighbours estimator. Because only ten strategy observations are available, the neighbourhood size must be reported explicitly and the resulting MI values should be interpreted as case-specific dependence estimates rather than population-level measures.

Step 8: Construction of the MI matrix and screening threshold

The pairwise MI estimates form the symmetric matrix

$$I = [I_{jk}]_{n \times n}, \quad I_{jk} = I_{kj}, \quad I_{jj} = 0. \quad (21)$$

The diagonal entries I_{jj} are set to zero by convention for redundancy screening.

Because the model was designed to retain ten criteria, the screening threshold is selected from the empirical distribution of the distinct off-diagonal MI values. Let η^* denote the smallest percentile that leaves exactly ten criteria after the elimination procedure in Step 10. The corresponding threshold θ_{MI} is

$$\theta_{\text{MI}} = \text{Percentile}_{\eta^*} (\{I_{jk} : 1 \leq j < k \leq n\}). \quad (22)$$

Only pairs satisfying $j < k$ are included when the empirical distribution is constructed, preventing symmetric entries from being counted twice. Criterion pairs satisfying $I_{jk} \geq \theta_{\text{MI}}$ are processed in descending order of MI. The selected percentile and resulting threshold are reported with the results and are determined independently of the subsequent MARCOS ranking.

Step 9: Calculation of criterion representativeness

The dependence of criterion D_j on the remainder of the criterion set is summarized as

$$r_j = \sum_{\substack{k=1 \\ k \neq j}}^n I_{jk}. \quad (23)$$

The quantity r_j is used here as a representativeness score rather than as a predictive-importance measure. It indicates the position of a criterion within the observed dependence structure and is used only in the redundancy-screening procedure.

Step 10: Removal of redundant criteria

For each criterion pair (j, k) flagged by the threshold rule, the criterion to be removed, denoted by D_{drop} , is the one with the smaller representativeness score:

$$D_{\text{drop}} = \arg \min_{D_\ell \in \{D_j, D_k\}} r_\ell. \quad (24)$$

Eq. (24) is applied only to criterion pairs that meet the MI threshold and the procedure stops when ten criteria remain. If $r_j = r_k$, the criterion with the clearer operational definition and the more direct connection to the policy question is retained, and the decision is reported explicitly.

Let $p = 10$ denote the number of retained criteria, and let v_ℓ denote the original column index of the ℓ th retained criterion. The retained criterion set D^* , its re-indexed criteria $\hat{D}_\ell$, and the corresponding original-index vector V are defined as

$$D^* = \{\hat{D}_1, \dots, \hat{D}_p\}, \quad \hat{D}_\ell = D_{v_\ell}, \quad V = (v_1, \dots, v_p). \quad (25)$$

The reduced score matrix X^* preserves the original criterion mapping:

$$X^* = [x_{i\ell}^*]_{m \times p}, \quad x_{i\ell}^* = x_{i, v_\ell}, \quad \ell = 1, \dots, p. \quad (26)$$

Eqs. (25) and (26) establish the selected-criterion notation used throughout the subsequent weighting and ranking stages.

Step 11: Restoration of criterion direction

The linguistic scale expresses favourable conditions for both benefit and cost criteria. The selected scores must therefore be transformed before the conventional benefit–cost treatment is applied. The transformed value is

$$y_{i\ell} = \begin{cases} \frac{x_{i\ell}^* + 1}{2}, & \hat{D}_\ell \text{ is a benefit criterion,} \\ \frac{1 - x_{i\ell}^*}{2}, & \hat{D}_\ell \text{ is a cost criterion.} \end{cases} \quad (27)$$

Eq. (27) is intended to prevent a second preference reversal for cost criteria that were originally expressed using favourable linguistic labels. After this transformation, larger $y_{i\ell}$ values are desirable for benefit criteria, whereas larger values for cost criteria represent greater burdens and are therefore treated accordingly in the subsequent normalization.

For each retained criterion,

$$y_\ell^{\max} = \max_i y_{i\ell}, \quad y_\ell^{\min} = \min_i y_{i\ell}. \quad (28)$$

These values define the observed criterion limits used in the next normalization stage.

Step 12: LOPCOW normalization

LOPCOW requires the criteria to be expressed in a common favourable direction. The normalized value $z_{i\ell}$ is calculated as

$$z_{i\ell} = \begin{cases} \frac{y_{i\ell} - y_\ell^{\min}}{y_\ell^{\max} - y_\ell^{\min}}, & \hat{D}_\ell \text{ is a benefit criterion,} \\ \frac{y_\ell^{\max} - y_{i\ell}}{y_\ell^{\max} - y_\ell^{\min}}, & \hat{D}_\ell \text{ is a cost criterion.} \end{cases} \quad (29)$$

If $y_\ell^{\max} = y_\ell^{\min}$, criterion $\hat{D}_\ell$ contains no variation across the candidate strategies and is removed before Eq. (29) is evaluated.

Step 13: Calculation of LOPCOW information quantities

For each retained criterion $\hat{D}_\ell$, the mean normalized value $\bar{z}_\ell$ is

$$\bar{z}_\ell = \frac{1}{m} \sum_{i=1}^m z_{i\ell}. \quad (30)$$

The dispersion measure s_ℓ is

$$s_\ell = \sqrt{\frac{1}{m} \sum_{i=1}^m (z_{i\ell} - \bar{z}_\ell)^2}. \quad (31)$$

The quadratic magnitude q_ℓ is

$$q_\ell = \sqrt{\frac{1}{m} \sum_{i=1}^m z_{i\ell}^2}. \quad (32)$$

Eqs. (30)–(32) describe the centre, dispersion, and quadratic magnitude of each normalized criterion, respectively. The LOPCOW information quantity L_ℓ is then calculated as

$$L_\ell = \left| 100 \ln \left(\frac{q_\ell}{s_\ell + \varepsilon} \right) \right|, \quad \varepsilon > 0. \quad (33)$$

The small constant ε prevents division by zero and is reported as part of the computational settings.

Step 14: Calculation of LOPCOW objective weights

The LOPCOW objective weight o_ℓ assigned to retained criterion $\hat{D}_\ell$ is calculated as

$$o_\ell = \frac{L_\ell}{\sum_{k=1}^p L_k}, \quad \ell = 1, 2, \dots, p. \quad (34)$$

The resulting LOPCOW weights are non-negative and satisfy

$$\sum_{\ell=1}^p o_\ell = 1.$$

Step 15: Construction of the RANCOM comparison basis

RANCOM is applied using the LOPCOW objective-weight ordering as its comparison basis. For each retained criterion, the comparison-basis value ϖ_ℓ is defined as

$$\varpi_\ell = o_\ell, \quad \ell = 1, 2, \dots, p. \quad (35)$$

Eq. (35) establishes a direct dependence between the LOPCOW and RANCOM stages. The resulting RANCOM weights are therefore interpreted as a transformation of the LOPCOW ordering rather than as an independently elicited set of subjective expert preferences.

Step 16: Construction of the RANCOM comparison matrix

For retained criteria $\hat{D}_\ell$ and $\hat{D}_r$, the pairwise comparison entry $c_{\ell r}$ is defined as

$$c_{\ell r} = \begin{cases} 1, & \varpi_\ell > \varpi_r, \\ 0.5, & \varpi_\ell = \varpi_r, \\ 0, & \varpi_\ell < \varpi_r. \end{cases} \quad (36)$$

Eq. (36) assigns one comparison point to the criterion with the larger LOPCOW weight and divides the point equally when the two weights are identical. The resulting comparison matrix is $C = [c_{\ell r}]_{p \times p}$.

The row total t_ℓ for criterion $\hat{D}_\ell$ is calculated as

$$t_\ell = \sum_{r=1}^p c_{\ell r}. \quad (37)$$

Step 17: Calculation of RANCOM weights

Normalizing the row totals gives the RANCOM weight b_ℓ :

$$b_\ell = \frac{t_\ell}{\sum_{r=1}^p t_r}, \quad \ell = 1, 2, \dots, p. \quad (38)$$

The resulting weights are non-negative and satisfy

$$\sum_{\ell=1}^p b_\ell = 1.$$

The vector $b = (b_1, b_2, \dots, b_p)$ is referred to as the RANCOM weight vector.

Step 18: Combination of criterion weights

Let $\alpha \in [0, 1]$ control the relative contribution of the objective LOPCOW weights. The preliminary combined weight $\tilde{w}_\ell$ for retained criterion $\hat{D}_\ell$ is calculated as

$$\tilde{w}_\ell = \alpha o_\ell + (1 - \alpha) b_\ell. \quad (39)$$

The same value of α is applied to every retained criterion. The final normalized weight w_ℓ^* is calculated as

$$w_\ell^* = \frac{\tilde{w}_\ell}{\sum_{r=1}^p \tilde{w}_r}. \quad (40)$$

Eq. (40) removes any numerical drift from the unit-sum condition. The baseline model uses $\alpha = 0.50$, which assigns equal numerical influence to the LOPCOW and RANCOM weight vectors. Alternative values of α are examined in sensitivity analysis rather than selected after the ranking is known. Because the RANCOM vector is derived from the ordering of the LOPCOW weights rather than from an independent source of preference information, the influence of the mixing parameter α is examined explicitly. Sensitivity analysis is conducted for $\alpha = 0.00, 0.25, 0.50, 0.75$, and 1.00 , ranging from a purely RANCOM-based specification to a purely LOPCOW-based specification.

Step 19: Construction of ideal and anti-ideal profiles

For retained criterion $\hat{D}_\ell$, the ideal value y_ℓ^+ is defined as

$$y_\ell^+ = \begin{cases} \max_i y_{i\ell}, & \hat{D}_\ell \text{ is a benefit criterion,} \\ \min_i y_{i\ell}, & \hat{D}_\ell \text{ is a cost criterion.} \end{cases} \quad (41)$$

The corresponding anti-ideal value y_ℓ^- is defined as

$$y_\ell^- = \begin{cases} \min_i y_{i\ell}, & \hat{D}_\ell \text{ is a benefit criterion,} \\ \max_i y_{i\ell}, & \hat{D}_\ell \text{ is a cost criterion.} \end{cases} \quad (42)$$

Eqs. (41) and (42) define the two reference profiles against which the candidate governance strategies are evaluated. The two reference rows are appended to the transformed decision matrix $Y = [y_{i\ell}]_{m \times p}$.

Step 20: MARCOS normalization

Each ordinary and reference row is normalized relative to the ideal profile to obtain the normalized value $n_{i\ell}$:

$$n_{i\ell} = \begin{cases} \frac{y_{i\ell}}{y_\ell^+}, & \hat{D}_\ell \text{ is a benefit criterion,} \\ \frac{y_\ell^-}{y_{i\ell}}, & \hat{D}_\ell \text{ is a cost criterion.} \end{cases} \quad (43)$$

If a denominator in Eq. (43) is zero, the same reported constant ε used in Eq. (33) is added to that denominator.

Step 21: Weighted normalized matrix and profile totals

The weighted normalized value $v_{i\ell}$ is calculated as

$$v_{i\ell} = n_{i\ell} w_\ell^*. \quad (44)$$

The weighted total S_i for strategy G_i , together with the ideal and anti-ideal reference totals S^+ and S^- , is calculated as follows, where v_ℓ^+ and v_ℓ^- denote the weighted normalized values of the ideal and anti-ideal reference profiles, respectively:

$$S_i = \sum_{\ell=1}^p v_{i\ell}, \quad S^+ = \sum_{\ell=1}^p v_\ell^+, \quad S^- = \sum_{\ell=1}^p v_\ell^-. \quad (45)$$

All three quantities in Eq. (45) are derived from the same weighted normalized matrix.

Step 22: Calculation of MARCOS utility degrees

The utility degrees K_i^- and K_i^+ of strategy G_i relative to the anti-ideal and ideal profiles, respectively, are calculated as

$$K_i^- = \frac{S_i}{S^-}, \quad K_i^+ = \frac{S_i}{S^+}. \quad (46)$$

These values locate each candidate strategy relative to the two reference profiles. The corresponding utility components f_i^- and f_i^+ are calculated as

$$f_i^- = \frac{K_i^+}{K_i^+ + K_i^-}, \quad f_i^+ = \frac{K_i^-}{K_i^+ + K_i^-}. \quad (47)$$

The two terms in Eq. (47) are subsequently incorporated into the final appraisal function.

Step 23: MARCOS appraisal score and ranking

The MARCOS appraisal score A_i of strategy G_i is calculated as

$$A_i = \frac{K_i^+ + K_i^-}{1 + \frac{1 - f_i^+}{f_i^+} + \frac{1 - f_i^-}{f_i^-}}. \quad (48)$$

Eq. (48) assigns one comparable appraisal value to each candidate strategy. Strategies are ranked in descending order of A_i . Accordingly,

$$A_i > A_k \Rightarrow G_i \succ G_k. \quad (49)$$

The highest-ranked alternative is interpreted only within the constructed assessment matrix and the stated parameter settings. The ranking therefore identifies the preferred alternative under the specified decision scenario rather than establishing a universally optimal digital-governance policy.

Table 6 summarizes the notation used throughout the proposed decision-support framework.

Table 6. Main symbols used in the methodology

Symbol	Description	Symbol	Description
G_i	The i th digital governance strategy	D_j	The j th original evaluation criterion
E_h	The h th assessment profile	$Z_{ij}^{(h)}$	Picture fuzzy evaluation assigned by E_h
w_h	Softmax weight of assessment profile E_h	$\bar{Z}_{ij}$	Aggregated picture fuzzy evaluation
x_{ij}	Crisp score of G_i under D_j	I_{jk}	MI between D_j and D_k
θ_{MI}	MI redundancy threshold	r_j	Representativeness score of criterion D_j
D^*	Set of retained criteria	$\hat{D}_\ell$	The ℓ th retained criterion
X^*	Reduced score matrix after criterion selection	$z_{i\ell}$	LOPCOW-normalized value
o_ℓ	LOPCOW objective weight of $\hat{D}_\ell$	$c_{\ell r}$	RANCOM comparison value between retained criteria ℓ and r
b_ℓ	RANCOM weight of $\hat{D}_\ell$	w_ℓ^*	Final combined weight of $\hat{D}_\ell$
y_ℓ^+	Ideal value of the ℓ th retained criterion	y_ℓ^-	Anti-ideal value of the ℓ th retained criterion
S_i	Total weighted MARCOS score of G_i	K_i^+, K_i^-	MARCOS utility degrees of G_i
A_i	Final MARCOS appraisal score	$\succ$	Strict preference relation between governance strategies

Note: MI = mutual information; LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; RANCOM = Ranking Comparison; MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution.

3.2 Statement of the Problem

Indonesian public authorities face multiple and often conflicting pressures when planning future digital public services. Residents need clear and reliable access to routine administrative services, yet access to stable connectivity, suitable digital devices, and physical government offices varies substantially across locations. Public agencies must also determine how administrative records should be shared, how service requests should be processed, and where responsibility should remain when automated or AI-assisted systems contribute to public decisions. Research on Indonesian digital government links successful service implementation to organizational capacity, infrastructure, transparency, and accountability [43]. Evidence from the Jakarta Kini application further highlights the importance of trust, perceived risk, service quality, and residents' ability to use digital service channels [3].

These concerns arise simultaneously when public authorities decide how limited resources should be allocated among competing digital-governance strategies. The alternatives considered in this study range from a unified public-service portal and mobile-first services for remote communities to predictive service management, urban digital-twin planning, citizen-participation platforms, inter-agency data exchange, and climate-responsive early-warning systems. These strategies do not address the same public-service problem in the same way. A shared portal primarily reduces fragmentation in finding and completing administrative transactions. A citizen-participation platform creates an institutional channel through which residents can express preferences and follow public responses. A data-exchange platform focuses instead on coordination and information flows across government agencies. An integrated governance ecosystem has a wider functional scope than a stand-alone e-government application but also imposes substantially greater requirements for institutional coordination and implementation.

The fit among proposed digital tools, existing administrative processes, data resources, and organizational structures has been identified as an important condition for AI adoption in Indonesian public institutions [46]. The decision criteria used in the present study therefore include both anticipated public-service benefits and the burdens imposed on public authorities and residents. Benefit criteria include public-service accessibility, service-delivery efficiency, citizen trust, psychological acceptance, digital inclusion, transparency and accountability, algorithmic fairness, and climate and environmental resilience. Cost criteria include initial and recurring expenditure, data-integration cost, implementation time, technological and administrative complexity, vendor dependence, and privacy, surveillance, and data-misuse risk.

A higher value indicates a more favourable outcome for a benefit criterion, whereas lower underlying burdens are preferable for cost criteria. This distinction is preserved explicitly throughout the transformation, normalization, weighting, and ranking stages. No single criterion is sufficient to determine the preferred strategy. Mobile access, for example, may reduce the need for residents to travel to administrative offices, but its effectiveness remains dependent on network availability, device access, digital literacy, and the availability of assisted service channels. Cross-agency data sharing may reduce repeated paperwork and improve coordination but simultaneously increases the need for clearly defined data-access rights and accountability mechanisms. Predictive analytics may assist cities in anticipating service demand and allocating personnel, yet their outputs require careful review when the underlying administrative records are incomplete or unrepresentative.

Research on digital innovation in Indonesia has examined infrastructure, financial resources, innovation adoption,

and service-related outcomes [7], whereas research on AI-enabled governance in Makassar has emphasized the actors and institutional relationships involved in public-service delivery [47]. The present study brings these concerns into a common decision-support framework rather than converting the empirical findings of those studies directly into numerical scores for the candidate alternatives.

The assessment matrices used in the case study were constructed to demonstrate the application of the proposed methodology. They should therefore be interpreted as illustrative decision inputs rather than as observed evaluations obtained from Indonesian citizens, municipal officials, or implemented public-service systems. Positive, neutral, and negative components are preserved until the group assessment has been formed, after which the decision matrix is transformed for criterion screening, weighting, and ranking.

The analytical question addressed by the case study is therefore conditional: given the stated assessments and parameter settings, how are the ten candidate digital urban governance strategies ranked, and is the separation among the leading alternatives sufficiently large to support a clear preference within this illustrative decision setting? The resulting ordering may inform the design of a future local assessment, but an actual investment decision would require locally observed service information, realistic financial and implementation estimates, and assessments obtained from the public institutions and residents affected by the proposed service.

The case study compares ten possible directions for future digital public-service development in Indonesia. Each alternative represents a strategy that a public authority could potentially prioritize rather than a project already implemented at national scale. Some alternatives address relatively specific service challenges, such as remote-island access or flood early warning, whereas others require coordination across multiple agencies. The cited studies provide substantive context for defining the alternatives, while the advantages and limitations summarized in Table 7 are used to frame the decision problem rather than to represent measured performance outcomes.

Table 7. Digital urban governance alternatives for the Indonesian case study

Strategy	Description	Public-Service Challenge Addressed	Typical Applications	Main Advantages	Main Limitations
Unified national–municipal digital public-service portal (G_1) [43]	A common entry point through which residents can locate services, submit requests, and monitor their progress across different levels of government.	Reduces fragmentation across service channels and repeated administrative steps in routine applications.	Permits, certificates, complaints, and service-status enquiries.	Can simplify service discovery and reduce repeated submission of the same information.	Requires agencies to align procedures, data fields, and responsibility for unresolved requests.
AI-driven predictive public-service management (G_2) [45]	Uses administrative data and predictive models to support service planning and the allocation of staff or other resources.	Addresses delays that arise when service demand changes faster than conventional planning cycles.	Demand forecasting, request triage, and identification of service backlogs.	Can help public agencies anticipate workload and direct resources toward services under pressure.	Performance depends on data quality; incomplete or unrepresentative records may lead to poor decisions.
Urban digital-twin planning and governance platform (G_3) [52]	Links digital representations of urban assets with data used to examine planning and management scenarios.	Addresses the difficulty of anticipating how proposed changes to urban infrastructure may affect interconnected city services.	Land-use scenarios, transport planning, and infrastructure maintenance.	Allows planners to evaluate possible consequences before committing resources.	Requires reliable spatial data, technical expertise, regular updates, and cooperation from asset owners.
Smart citizen-participation and electronic consultation system (G_4) [47]	Provides digital channels through which residents can comment on proposals, report local concerns, and follow government responses.	Addresses limited opportunities for incorporating public preferences into urban service planning and governance.	Public consultation, participatory planning, and neighbourhood issue reporting.	Makes citizen comments and government responses easier to record, review, and trace.	Participation may favour residents with better connectivity unless adequate offline channels are also maintained.

(continued)

(continued)

Privacy-preserving urban data-exchange platform (G_5) [43]	Establishes institutional rules and technical controls for sharing service-related data among authorised public agencies.	Addresses repeated data collection while maintaining privacy, accountability, and appropriate access controls.	Cross-agency referrals, record verification, and coordinated service requests.	Can reduce duplicate information submission and clarify access rights across agencies.	Requires agreement on data quality, access rights, retention, and institutional responsibility for misuse.
Integrated digital identity for public-service access (G_6) [53]	Uses digital identity mechanisms to verify residents when they access eligible public services.	Addresses repeated identity verification across fragmented service channels.	Account access, application verification, and retrieval of personal service records.	Can simplify access for residents who use multiple government services.	Enrollment or authentication errors may exclude users; assisted and non-digital access channels remain necessary.
Mobile-first services for remote and island communities (G_7) [54]	Designs essential public-service transactions around mobile devices and connectivity conditions outside well-served urban centres.	Addresses travel burdens, uneven access to government offices, and limited fixed internet connectivity.	Mobile applications, low-bandwidth forms, and assisted service requests.	Can reduce the need for physical travel among residents who are able to use mobile channels.	Effectiveness depends on coverage, device access, digital skills, and the availability of in-person assistance.
Climate-responsive urban governance and early-warning system (G_8) [55]	Integrates hazard monitoring, public warning mechanisms, and the agencies responsible for emergency response.	Addresses delays between hazard detection and the communication of actionable information to affected residents.	Flood alerts, local risk information, and coordinated emergency response.	Can support earlier warning and improve the traceability of institutional responses.	Depends on sensor maintenance, reliable communications, and messages that residents can understand and act upon.
Blockchain-based transparent public-administration system (G_9) [56]	Uses shared, auditable records for selected administrative transactions and document workflows.	Addresses uncertainty over when records were changed and which office handled a particular administrative action.	Document tracking, workflow logs, and verification of administrative records.	Can improve traceability in processes that require a shared audit trail.	Requires careful integration and protection of sensitive data; auditability alone does not prevent unfair decisions.
Integrated intelligent urban governance ecosystem (G_{10}) [46]	Connects public-service channels, agency data, analytical tools, and governance arrangements within a broader urban-management programme.	Addresses fragmentation across services that cannot be resolved by modifying a single application.	Coordinated service delivery, planning dashboards, and cross-agency case management.	Can help agencies examine urban service problems across administrative boundaries.	Its broader scope increases requirements for financing, coordination, technical maintenance, and institutional oversight.

Note: G_i = the i th digital governance strategy ($i = 1, \dots, 10$).

The evaluation criteria similarly represent expected service benefits and implementation burdens. The first eight criteria are benefit criteria, for which a higher assessment represents a more favourable expected outcome. The remaining seven are cost criteria, for which a lower underlying burden is preferred. Here, cost is interpreted broadly to include financial expenditure, implementation time, organizational and technological complexity, dependence on external providers, and privacy-related risk. Table 8 provides the operational definition of each criterion before the decision matrix is introduced.

The direction of every criterion is fixed before normalization and ranking. In particular, a high value for D_{15} represents greater privacy, surveillance, and data-misuse risk and must not be interpreted as stronger privacy protection. The cited literature supports the selection and conceptual definition of the criteria; it does not provide the numerical

assessments used in the constructed case study.

Table 8. Criteria for assessing digital urban governance strategies

Category	Criterion	Description	Interpretation of a Higher Value	Type
Service access	Public-service accessibility, D_1 [3]	Ease with which residents can reach and complete public services across urban, remote, and island communities.	More residents can obtain the service with less travel or procedural difficulty.	Benefit
Service performance	Service-delivery efficiency, D_2 [7]	Expected effect on processing time, administrative delay, paperwork, and repeated procedures.	Greater expected reduction in processing time or duplicated administrative work.	Benefit
Public confidence	Citizen trust, D_3 [3]	Public confidence that a service operates reliably and that the responsible authority handles requests fairly.	Greater willingness to rely on the service for official transactions.	Benefit
User response	Psychological acceptance, D_4 [3]	Residents' willingness and comfort in using automated or data-driven public-service channels.	Fewer perceived barriers to initial and continued use.	Benefit
Equitable access	Digital inclusion, D_5 [45]	Capacity to serve residents who face barriers related to income, age, disability, geographic location, or digital skills.	Fewer residents remain dependent on inaccessible service channels.	Benefit
Administrative governance	Transparency and accountability, D_6 [43]	Ability to trace public-service actions and identify institutional responsibility for decisions, errors, or corrections.	Administrative procedures become easier to examine, explain, and contest.	Benefit
AI governance	Algorithmic fairness, D_7 [57]	Extent to which automated processes avoid unequal treatment across relevant population groups.	Lower expected risk of discriminatory outcomes under the stated assessment.	Benefit
Environmental response	Climate and environmental resilience, D_8 [55]	Expected contribution to climate adaptation, hazard information, emergency response, or efficient resource use.	Stronger capacity to maintain or adapt public services under environmental stress.	Benefit
Capital expenditure	Initial implementation cost, D_9 [7]	Upfront expenditure required for infrastructure, software, equipment, staff training, and deployment.	Greater initial financial burden on the public budget.	Cost
Recurring expenditure	Operation and maintenance cost, D_{10} [7]	Continuing expenditure required for technical support, cybersecurity, system upgrades, storage, and routine operation.	Greater ongoing resource requirements to keep the service operational.	Cost
System integration	Data-integration cost, D_{11} [58]	Financial and technical effort required to connect existing databases and agency information systems.	Greater difficulty and expense associated with cross-agency deployment.	Cost
Delivery schedule	Implementation time, D_{12} [46]	Time required to design, test, deploy, and establish routine use of a digital governance strategy.	Longer delay before residents can obtain the intended service benefits.	Cost
Organisational demands	Technological and administrative complexity, D_{13} [43]	Difficulty associated with system management, staff preparation, regulatory coordination, and cooperation across agencies.	Greater demands on the institutions responsible for implementation and operation.	Cost
Supplier dependence	Vendor dependence and technological lock-in, D_{14} [59]	Risk that future operation, modification, or replacement depends heavily on a single provider, proprietary technology, or scarce expertise.	Greater difficulty in switching providers or adapting the system over time.	Cost
Data protection	Privacy, surveillance, and data-misuse risk, D_{15} [49]	Risk of unauthorised use of personal data, excessive monitoring, or inappropriate profiling.	Greater potential harm and a stronger need for safeguards, accountability, and oversight.	Cost

Note: D_j = the j th original evaluation criterion ($j = 1, \dots, 15$).

4 Results and Robustness Analysis

4.1 Baseline Evaluation and Ranking

The baseline calculation used $\tau = 1$, $\kappa = 2$, $k = 3$, $\varepsilon = 10^{-12}$, and $\alpha = 0.50$.

Numerical values reported in the tables and figures are rounded for presentation, whereas unrounded values were retained throughout all subsequent calculations. Although “acceptable under existing conditions” (AEC) and “unsuitable for urban governance implementation” (UGI) remained valid categories within the eight-level linguistic scale, neither occurred in the constructed decision matrices.

Step 1: Linguistic assessment matrices

Table 9. Constructed linguistic evaluations for the four role-based assessment profiles

Assessment Profile	Strategy	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}
E_1	G_1	HSP	HSP	HSP	HSP	HSP	SMC	SMC	MSI	MSI	SMC	MSI	SMC	MSI	SMC	SMC
	G_2	SMC	FSG	MSI	MSI	MSI	SMC	MSI	HSP	LSP	LSP	MSI	MSI	LSP	LSP	MSI
	G_3	MSI	HSP	SMC	MSI	MSI	HSP	SMC	FSG	LSP	LSP	LSP	LSP	LSP	PSC	SMC
	G_4	HSP	SMC	FSG	FSG	HSP	FSG	HSP	MSI	HSP	HSP	SMC	HSP	SMC	HSP	HSP
	G_5	SMC	SMC	HSP	HSP	MSI	HSP	FSG	SMC	MSI	MSI	LSP	MSI	LSP	SMC	FSG
	G_6	HSP	HSP	SMC	SMC	HSP	HSP	SMC	MSI	MSI	SMC	LSP	SMC	MSI	LSP	SMC
	G_7	FSG	HSP	HSP	HSP	FSG	SMC	HSP	SMC	HSP	HSP	HSP	HSP	SMC	HSP	HSP
	G_8	SMC	HSP	HSP	SMC	SMC	HSP	SMC	FSG	MSI	MSI	MSI	SMC	MSI	SMC	HSP
	G_9	MSI	SMC	MSI	MSI	LSP	FSG	SMC	SMC	LSP	LSP	LSP	LSP	PSC	PSC	MSI
	G_{10}	FSG	FSG	HSP	HSP	HSP	FSG	HSP	FSG	PSC	LSP	PSC	LSP	PSC	LSP	SMC
E_2	G_1	HSP	FSG	SMC	HSP	SMC	HSP	SMC	MSI	MSI	SMC	MSI	SMC	MSI	MSI	SMC
	G_2	HSP	FSG	MSI	MSI	MSI	SMC	SMC	HSP	LSP	LSP	LSP	MSI	PSC	LSP	LSP
	G_3	MSI	HSP	MSI	MSI	LSP	SMC	HSP	FSG	PSC	LSP	PSC	LSP	PSC	PSC	SMC
	G_4	HSP	SMC	HSP	HSP	HSP	FSG	HSP	MSI	HSP	HSP	SMC	HSP	SMC	HSP	HSP
	G_5	SMC	HSP	FSG	HSP	SMC	HSP	FSG	SMC	MSI	MSI	LSP	MSI	LSP	SMC	FSG
	G_6	HSP	FSG	SMC	SMC	HSP	HSP	HSP	MSI	MSI	SMC	LSP	SMC	MSI	LSP	HSP
	G_7	FSG	HSP	HSP	HSP	FSG	SMC	SMC	SMC	HSP	HSP	HSP	HSP	SMC	HSP	HSP
	G_8	SMC	HSP	SMC	SMC	SMC	HSP	SMC	FSG	MSI	MSI	MSI	SMC	MSI	SMC	HSP
	G_9	MSI	HSP	SMC	MSI	LSP	FSG	HSP	MSI	LSP	LSP	LSP	LSP	PSC	LSP	SMC
	G_{10}	FSG	FSG	HSP	HSP	HSP	FSG	FSG	HSP	PSC	LSP	PSC	PSC	PSC	LSP	MSI
E_3	G_1	HSP	HSP	HSP	HSP	HSP	SMC	HSP	MSI	SMC	SMC	MSI	SMC	MSI	HSP	HSP
	G_2	SMC	FSG	SMC	MSI	MSI	SMC	MSI	HSP	LSP	LSP	MSI	MSI	LSP	LSP	LSP
	G_3	MSI	HSP	MSI	LSP	LSP	SMC	SMC	FSG	LSP	LSP	LSP	LSP	PSC	PSC	MSI
	G_4	FSG	SMC	FSG	FSG	HSP	FSG	HSP	MSI	HSP	HSP	HSP	HSP	SMC	HSP	FSG
	G_5	SMC	SMC	FSG	HSP	SMC	HSP	FSG	SMC	MSI	MSI	LSP	MSI	LSP	SMC	FSG
	G_6	HSP	HSP	SMC	SMC	HSP	HSP	SMC	MSI	MSI	SMC	LSP	SMC	MSI	LSP	HSP
	G_7	FSG	HSP	FSG	HSP	FSG	SMC	HSP	SMC	HSP	HSP	HSP	FSG	SMC	HSP	HSP
	G_8	SMC	HSP	HSP	HSP	SMC	HSP	SMC	FSG	MSI	MSI	MSI	SMC	MSI	SMC	HSP
	G_9	MSI	SMC	MSI	LSP	LSP	FSG	SMC	MSI	LSP	LSP	LSP	LSP	PSC	PSC	MSI
	G_{10}	FSG	FSG	HSP	HSP	HSP	FSG	HSP	FSG	PSC	LSP	PSC	LSP	PSC	LSP	MSI
E_4	G_1	HSP	HSP	SMC	SMC	HSP	SMC	SMC	SMC	MSI	SMC	MSI	SMC	MSI	SMC	HSP
	G_2	SMC	FSG	MSI	MSI	MSI	SMC	MSI	HSP	LSP	LSP	MSI	MSI	LSP	LSP	MSI
	G_3	MSI	HSP	MSI	MSI	LSP	HSP	SMC	FSG	LSP	LSP	LSP	LSP	LSP	PSC	SMC
	G_4	HSP	SMC	HSP	HSP	HSP	FSG	HSP	SMC	HSP	HSP	SMC	HSP	SMC	HSP	HSP
	G_5	SMC	SMC	HSP	HSP	MSI	HSP	FSG	SMC	MSI	MSI	LSP	MSI	LSP	SMC	FSG
	G_6	HSP	HSP	SMC	SMC	HSP	HSP	SMC	MSI	MSI	SMC	LSP	SMC	MSI	LSP	HSP
	G_7	FSG	HSP	HSP	HSP	FSG	SMC	HSP	HSP	HSP	HSP	HSP	HSP	SMC	HSP	HSP
	G_8	HSP	HSP	SMC	SMC	SMC	HSP	SMC	FSG	MSI	MSI	MSI	SMC	MSI	SMC	HSP
	G_9	MSI	SMC	MSI	MSI	LSP	FSG	SMC	SMC	LSP	LSP	LSP	LSP	PSC	PSC	MSI
	G_{10}	FSG	FSG	HSP	HSP	HSP	FSG	HSP	FSG	PSC	LSP	PSC	LSP	PSC	LSP	SMC

Note: E_h = the h th assessment profile ($h = 1, \dots, 4$); G_i = the i th digital governance strategy ($i = 1, \dots, 10$); D_j = the j th original evaluation criterion ($j = 1, \dots, 15$); FSG = fully suitable for future urban governance; HSP = highly suitable for public-service transformation; SMC = suitable with minor implementation concerns; MSI = moderately suitable for urban implementation; LSP = limited suitability for public-service delivery; PSC = poorly suited to current institutional capacity.

Eq. (7) produced four separate 10×15 linguistic assessment matrices corresponding to the four role-based assessment profiles. Table 9 reports the complete constructed evaluations, with the assessment profile identified in the first column. Each profile contains 150 strategy–criterion assessments, resulting in 600 illustrative entries in total. Input validation confirmed that every entry belonged to the predefined eight-level linguistic scale. These entries were constructed specifically for the methodological case study and should not be interpreted as responses collected from recruited experts or other human participants.

Step 2: Conversion to PiFNs

The linguistic evaluations were converted into PiFNs according to Eq. (8) and the scale defined in Table 4. Table 10 reports the positive, neutral, and negative membership degrees associated with each linguistic category, together with the total assigned degree and corresponding refusal degree. Every category satisfied the constraints specified in Eqs. (9) and (10).

Table 10. Picture fuzzy degrees assigned to the linguistic labels

Label	a	b	c	$a + b + c$	Refusal Degree d
FSG	0.90	0.03	0.03	0.96	0.04
HSP	0.82	0.06	0.07	0.95	0.05
SMC	0.74	0.09	0.12	0.95	0.05
MSI	0.65	0.13	0.17	0.95	0.05
AEC	0.55	0.18	0.22	0.95	0.05
LSP	0.45	0.22	0.28	0.95	0.05
PSC	0.34	0.27	0.34	0.95	0.05
UGI	0.22	0.30	0.43	0.95	0.05

Note: FSG = fully suitable for future urban governance; HSP = highly suitable for public-service transformation; SMC = suitable with minor implementation concerns; MSI = moderately suitable for urban implementation; AEC = acceptable under existing conditions; LSP = limited suitability for public-service delivery; PSC = poorly suited to current institutional capacity; UGI = unsuitable for urban governance implementation; a = positive membership degree; b = neutral membership degree; c = negative membership degree; d = refusal degree; $a + b + c$ = sum of the positive, neutral, and negative membership degrees.

Steps 3–4: Profile-specific competence specification and weights

Applying the score function in Eq. (11) produced, for example, a score of 0.84 for the “fully suitable for future urban governance” (FSG) category and 0.69 for the “highly suitable for public-service transformation” (HSP) category. The four competence dimensions were assigned equal importance, $\rho_r = 0.25$.

Table 11 reports the constructed competence specifications used to differentiate the four role-based assessment profiles and the resulting profile scores. These values are methodological inputs for demonstrating the weighting procedure and are not empirical measurements of the qualifications of recruited individuals.

Table 11. Constructed competence specifications and resulting assessment-profile scores

Assessment Profile	Experience	Problem Knowledge	Technical Knowledge	Context Knowledge	Total Score e_h
E_1	FSG	HSP	HSP	FSG	0.7650
E_2	HSP	FSG	FSG	HSP	0.7650
E_3	HSP	HSP	FSG	FSG	0.7650
E_4	HSP	HSP	HSP	FSG	0.7275

Note: E_h = the h th assessment profile ($h = 1, 2, 3, 4$); e_h = overall competence score of the h th assessment profile; FSG = fully suitable for future urban governance; HSP = highly suitable for public-service transformation.

Table 12. Assessment-profile competence scores and Softmax weights

Assessment Profile	Competence Score e_h	Softmax Weight w_h
E_1	0.7650	0.252322
E_2	0.7650	0.252322
E_3	0.7650	0.252322
E_4	0.7275	0.243035
Total	—	1.000000 (before rounding)

Note: E_h = the h th assessment profile ($h = 1, 2, 3, 4$); “—” indicates not applicable.

Eq. (13) transformed the competence scores into Softmax weights. The results are reported in Table 12 and illustrated in Figure 1. Assessment profiles E_1 , E_2 , and E_3 received identical weights because their constructed competence scores were equal, whereas profile E_4 received a slightly lower weight.

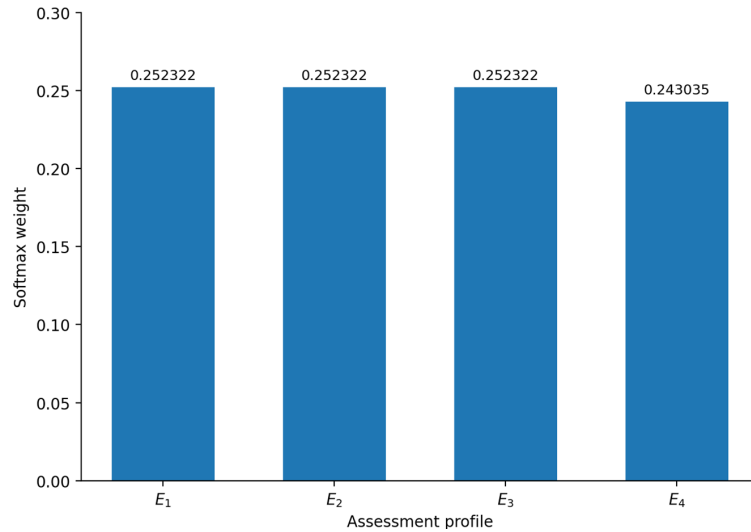


Figure 1. Softmax weights assigned to the four assessment profiles
Note: E_h is the h th assessment profile ($h = 1, 2, 3, 4$).

Step 5: Aggregated picture fuzzy decision matrix

Eqs. (14)–(16) were applied to all 150 strategy–criterion positions. The resulting aggregated picture fuzzy decision matrix is reported in Table 13. Each matrix element is an ordered triple of the form $\langle \bar{a}_{ij}, \bar{b}_{ij}, \bar{c}_{ij} \rangle$, where the three components represent the aggregated positive, neutral, and negative membership degrees for strategy G_i under criterion D_j .

Table 13. Aggregated picture fuzzy decision matrix

Strategy	D_1	D_2	D_3	D_4	D_5
G_1	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8571, 0.0451, 0.0627 \rangle$	$\langle 0.7919, 0.0704, 0.0990 \rangle$	$\langle 0.8079, 0.0645, 0.0858 \rangle$	$\langle 0.8073, 0.0647, 0.0863 \rangle$
G_2	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.6825, 0.1149, 0.1596 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$
G_3	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.6825, 0.1149, 0.1596 \rangle$	$\langle 0.6237, 0.1418, 0.2087 \rangle$	$\langle 0.5393, 0.1802, 0.2602 \rangle$
G_4	$\langle 0.8571, 0.0451, 0.0627 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.8773, 0.0378, 0.0542 \rangle$	$\langle 0.8773, 0.0378, 0.0542 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$
G_5	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.8773, 0.0378, 0.0542 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.7065, 0.1043, 0.1480 \rangle$
G_6	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8571, 0.0451, 0.0627 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$
G_7	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8571, 0.0451, 0.0627 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.9000, 0.0300, 0.0300 \rangle$
G_8	$\langle 0.7697, 0.0787, 0.1106 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.7919, 0.0704, 0.0990 \rangle$	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$
G_9	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.6825, 0.1149, 0.1596 \rangle$	$\langle 0.6237, 0.1418, 0.2087 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$
G_{10}	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$
Strategy	D_6	D_7	D_8	D_9	D_{10}
G_1	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.6815, 0.1154, 0.1600 \rangle$	$\langle 0.6825, 0.1149, 0.1596 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$
G_2	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.6825, 0.1149, 0.1596 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$
G_3	$\langle 0.7913, 0.0706, 0.0994 \rangle$	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.4296, 0.2297, 0.2979 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$
G_4	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.6815, 0.1154, 0.1600 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$
G_5	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$
G_6	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$
G_7	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.8073, 0.0647, 0.0863 \rangle$	$\langle 0.7697, 0.0787, 0.1106 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$
G_8	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$
G_9	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.7058, 0.1046, 0.1484 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$
G_{10}	$\langle 0.9000, 0.0300, 0.0300 \rangle$	$\langle 0.8571, 0.0451, 0.0627 \rangle$	$\langle 0.8903, 0.0333, 0.0443 \rangle$	$\langle 0.3400, 0.2700, 0.3400 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$
Strategy	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}
G_1	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.7608, 0.0824, 0.1272 \rangle$	$\langle 0.7913, 0.0706, 0.0994 \rangle$
G_2	$\langle 0.6237, 0.1418, 0.2087 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.4296, 0.2297, 0.2979 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.5886, 0.1577, 0.2376 \rangle$
G_3	$\langle 0.4296, 0.2297, 0.2979 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.4056, 0.2410, 0.3137 \rangle$	$\langle 0.3400, 0.2700, 0.3400 \rangle$	$\langle 0.7247, 0.0965, 0.1353 \rangle$
G_4	$\langle 0.7707, 0.0784, 0.1102 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8571, 0.0451, 0.0627 \rangle$
G_5	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.9000, 0.0300, 0.0300 \rangle$
G_6	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.8073, 0.0647, 0.0863 \rangle$
G_7	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8571, 0.0451, 0.0627 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$
G_8	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.6500, 0.1300, 0.1700 \rangle$	$\langle 0.7400, 0.0900, 0.1200 \rangle$	$\langle 0.8200, 0.0600, 0.0700 \rangle$
G_9	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.3400, 0.2700, 0.3400 \rangle$	$\langle 0.3774, 0.2539, 0.3272 \rangle$	$\langle 0.6825, 0.1149, 0.1596 \rangle$
G_{10}	$\langle 0.3400, 0.2700, 0.3400 \rangle$	$\langle 0.4296, 0.2297, 0.2979 \rangle$	$\langle 0.3400, 0.2700, 0.3400 \rangle$	$\langle 0.4500, 0.2200, 0.2800 \rangle$	$\langle 0.7058, 0.1046, 0.1484 \rangle$

Note: G_i is the i th digital governance strategy ($i = 1, \dots, 10$); D_j is the j th original evaluation criterion ($j = 1, \dots, 15$).

The numerical entries are retained exactly as reported in the original calculation.

Step 6: Crisp decision matrix

Eq. (18) transformed each aggregated picture fuzzy assessment into a scalar score, producing the crisp decision matrix $X = [x_{ij}]_{m \times n}$ presented in Table 14. The scores ranged from -0.270 to 0.840 , and every criterion exhibited variation across the ten candidate strategies. No criterion was therefore removed at this stage because of a constant column.

Table 14. Crisp score matrix X derived from the aggregated evaluations

Strategy	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}
G_1	0.690	0.749	0.623	0.658	0.656	0.582	0.582	0.406	0.408	0.530	0.350	0.530	0.350	0.551	0.621
G_2	0.582	0.840	0.408	0.350	0.350	0.530	0.408	0.690	-0.050	-0.050	0.273	0.350	-0.098	-0.050	0.193
G_3	0.350	0.690	0.408	0.273	0.099	0.621	0.582	0.840	-0.098	-0.050	-0.098	-0.050	-0.149	-0.270	0.493
G_4	0.749	0.530	0.785	0.785	0.690	0.840	0.690	0.406	0.690	0.690	0.582	0.690	0.530	0.690	0.749
G_5	0.530	0.582	0.785	0.690	0.454	0.690	0.840	0.530	0.350	0.350	-0.050	0.350	-0.050	0.530	0.840
G_6	0.690	0.749	0.530	0.530	0.690	0.690	0.582	0.350	0.350	0.530	-0.050	0.530	0.350	-0.050	0.656
G_7	0.840	0.690	0.749	0.690	0.840	0.530	0.656	0.580	0.690	0.690	0.690	0.749	0.530	0.690	0.690
G_8	0.580	0.690	0.623	0.582	0.530	0.690	0.530	0.840	0.350	0.350	0.350	0.530	0.350	0.530	0.690
G_9	0.350	0.582	0.408	0.273	-0.050	0.840	0.582	0.453	-0.050	-0.050	-0.050	-0.050	-0.270	-0.204	0.408
G_{10}	0.840	0.840	0.690	0.690	0.690	0.840	0.749	0.813	-0.270	-0.050	-0.270	-0.098	-0.270	-0.050	0.453

Note: G_i = the i th digital governance strategy ($i = 1, \dots, 10$); D_j = the j th original evaluation criterion ($j = 1, \dots, 15$).

Steps 7–8: MI estimation and screening threshold

MI was estimated using Eq. (20) with $k = 3$ nearest neighbours. Given the limited number of strategy observations, the resulting MI values are interpreted as case-specific dependence measures rather than population-level estimates.

As shown in Table 15, the strongest estimated dependence occurred between D_9 and D_{10} , with $I_{9,10} = 0.828968$.

Table 15. Nine largest pairwise MI estimates

Rank	Criterion Pair	MI Estimate
1	D_9-D_{10}	0.828968
2	$D_{10}-D_{13}$	0.669683
3	D_9-D_{12}	0.665516
4	D_3-D_4	0.585635
5	$D_{12}-D_{13}$	0.577063
6	D_9-D_{14}	0.561349
7	D_9-D_{13}	0.555635
8	$D_{10}-D_{15}$	0.494683
9	D_5-D_{12}	0.461349

Note: MI = mutual information; D_j = the j th original evaluation criterion ($j = 1, \dots, 15$).

The diagonal terms were set to $I_{jj} = 0$. Eq. (22) yielded the screening threshold $\theta_{\text{MI}} = 0.555635$, corresponding to $\eta^* = 94.29\%$. The MI screening settings are summarized in Table 16. The threshold was applied to the unrounded MI estimates.

Table 16. MI screening settings

Quantity	Value
Distinct criterion pairs	105
Number of neighbours	3
Empirical cut-off position, η^*	94.29%
MI cut-off, θ_{MI}	0.555635
Target number of retained criteria	10

Note: MI = mutual information.

The complete pairwise MI matrix is presented in Table 17, with the corresponding heatmap shown in Figure 2.

Steps 9–10: Criterion representativeness and redundancy screening

Eq. (23) produced the representativeness score r_j for each criterion, as reported in Table 18. These values describe the position of each criterion within the observed dependence structure and are not criterion importance weights.

Applying Eq. (24) removed D_{10} , D_{12} , D_3 , D_{14} , and D_{13} , in that order. The original-index vector of the retained criteria was $V = (1, 2, 4, 5, 6, 7, 8, 9, 11, 15)$. The ten retained criteria and their original indices are presented in Table 19.

Table 17. Complete pairwise MI matrix

Criterion	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}
D_1	0.000	0.000	0.144	0.281	0.385	0.032	0.082	0.000	0.125	0.227	0.079	0.238	0.198	0.000	0.073
D_2	0.000	0.000	0.084	0.000	0.000	0.127	0.209	0.000	0.000	0.000	0.035	0.000	0.000	0.000	0.293
D_3	0.144	0.084	0.000	0.586	0.142	0.040	0.159	0.000	0.211	0.124	0.019	0.097	0.017	0.265	0.234
D_4	0.281	0.000	0.586	0.000	0.232	0.000	0.125	0.000	0.248	0.209	0.050	0.131	0.037	0.355	0.190
D_5	0.385	0.000	0.142	0.232	0.000	0.018	0.055	0.008	0.230	0.362	0.137	0.461	0.336	0.009	0.171
D_6	0.032	0.127	0.040	0.000	0.018	0.000	0.175	0.000	0.131	0.000	0.002	0.038	0.000	0.000	0.000
D_7	0.082	0.209	0.159	0.125	0.055	0.175	0.000	0.000	0.000	0.040	0.010	0.089	0.000	0.000	0.150
D_8	0.000	0.000	0.000	0.000	0.008	0.000	0.000	0.000	0.022	0.111	0.000	0.073	0.061	0.000	0.000
D_9	0.125	0.000	0.211	0.248	0.230	0.131	0.000	0.022	0.000	0.829	0.178	0.666	0.556	0.561	0.298
D_{10}	0.227	0.000	0.124	0.209	0.362	0.000	0.040	0.111	0.829	0.000	0.103	0.447	0.670	0.436	0.495
D_{11}	0.079	0.035	0.019	0.050	0.137	0.002	0.010	0.000	0.178	0.103	0.000	0.313	0.336	0.375	0.048
D_{12}	0.238	0.000	0.097	0.131	0.461	0.038	0.089	0.073	0.666	0.447	0.313	0.000	0.577	0.213	0.248
D_{13}	0.198	0.000	0.017	0.037	0.336	0.000	0.000	0.061	0.556	0.670	0.336	0.577	0.000	0.319	0.307
D_{14}	0.000	0.000	0.265	0.355	0.009	0.000	0.000	0.000	0.561	0.436	0.375	0.213	0.319	0.000	0.378
D_{15}	0.073	0.293	0.234	0.190	0.171	0.000	0.150	0.000	0.298	0.495	0.048	0.248	0.307	0.378	0.000

Note: MI = mutual information; D_j = the j th original evaluation criterion ($j = 1, \dots, 15$).

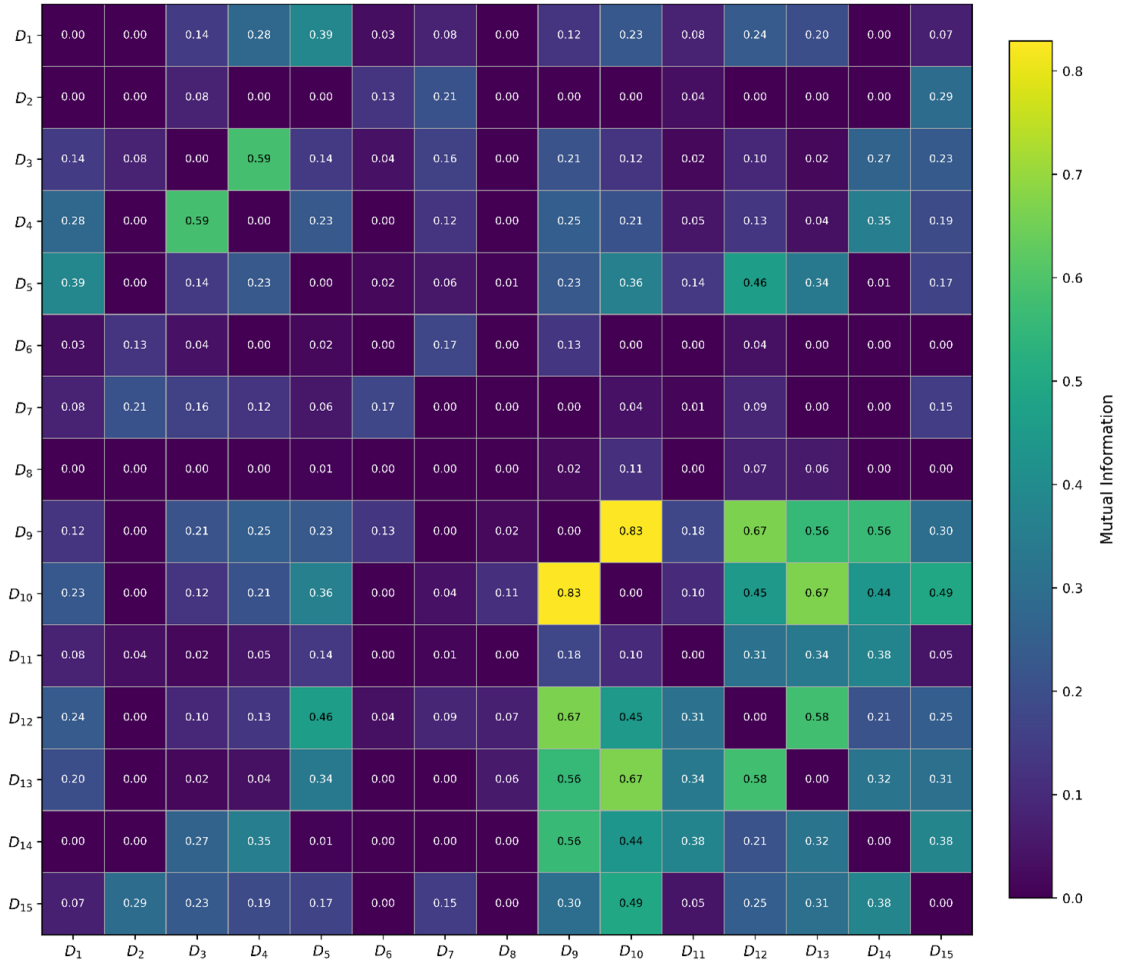


Figure 2. Heatmap of pairwise MI estimates among the original evaluation criteria

Note: MI = mutual information; D_j = the j th original evaluation criterion ($j = 1, \dots, 15$).

Table 18. Criterion representativeness scores used for screening

Criterion	Representativeness Score r_j
D_1	1.863095
D_2	0.747579
D_3	2.120873
D_4	2.443849
D_5	2.546190
D_6	0.561984
D_7	1.095000
D_8	0.274603
D_9	4.054167
D_{10}	4.052262
D_{11}	1.683730
D_{12}	3.591111
D_{13}	3.413413
D_{14}	2.910952
D_{15}	2.883730

Note: D_j = the j th original evaluation criterion ($j = 1, \dots, 15$).

Accordingly, $\hat{D}_\ell = D_{v_\ell}$ throughout the subsequent LOPCOW, RANCOM, and MARCOS calculations.

Table 19. Criteria retained after MI screening

ℓ	Selected Criterion	Criterion Name	Type
1	$\hat{D}_1 = D_1$	Public-service accessibility	Benefit
2	$\hat{D}_2 = D_2$	Service-delivery efficiency	Benefit
3	$\hat{D}_3 = D_4$	Psychological acceptance	Benefit
4	$\hat{D}_4 = D_5$	Digital inclusion	Benefit
5	$\hat{D}_5 = D_6$	Transparency and accountability	Benefit
6	$\hat{D}_6 = D_7$	Algorithmic fairness	Benefit
7	$\hat{D}_7 = D_8$	Climate and environmental resilience	Benefit
8	$\hat{D}_8 = D_9$	Initial implementation cost	Cost
9	$\hat{D}_9 = D_{11}$	Data-integration cost	Cost
10	$\hat{D}_{10} = D_{15}$	Privacy, surveillance, and data-misuse risk	Cost

Note: MI = mutual information; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$); D_j = the j th original evaluation criterion ($j = 1, \dots, 15$); Benefit = criterion for which a higher value is preferred; Cost = criterion for which a lower value is preferred.

Table 20. Column limits after restoring criterion direction

Selected Criterion	Minimum	Maximum
$\hat{D}_1$	0.675000	0.920000
$\hat{D}_2$	0.765000	0.920000
$\hat{D}_3$	0.636655	0.892665
$\hat{D}_4$	0.475000	0.920000
$\hat{D}_5$	0.765000	0.920000
$\hat{D}_6$	0.704005	0.920000
$\hat{D}_7$	0.675000	0.920000
$\hat{D}_8$	0.155000	0.635000
$\hat{D}_9$	0.155000	0.635000
$\hat{D}_{10}$	0.080000	0.403358

Note: $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$).

Step 11: Restoration of criterion direction

The retained criterion scores were transformed using Eq. (27) to restore the intended benefit–cost directions and

map the values to the interval $[0, 1]$. The resulting minimum and maximum values for each retained criterion are reported in Table 20.

Steps 12–14: LOPCOW normalization and objective weighting

Eq. (29) normalized every retained criterion to $[0, 1]$, with larger normalized values representing more favourable performance. Every retained criterion had a nonzero observed range, so normalization was defined for all ten criteria. The summary statistics of the normalized criteria are presented in Table 21, while the LOPCOW intermediate quantities and resulting objective weights are reported in Table 22.

The largest objective weight was $o_{10} = 0.130500$, corresponding to $\hat{D}_{10} = D_{15}$, or privacy, surveillance, and data-misuse risk.

Table 21. Summary statistics for the LOPCOW-normalized criteria

Selected Criterion	Minimum	Maximum	Mean
$\hat{D}_1$	0	1	0.551391
$\hat{D}_2$	0	1	0.529896
$\hat{D}_3$	0	1	0.544625
$\hat{D}_4$	0	1	0.612317
$\hat{D}_5$	0	1	0.501076
$\hat{D}_6$	0	1	0.491220
$\hat{D}_7$	0	1	0.491459
$\hat{D}_8$	0	1	0.528128
$\hat{D}_9$	0	1	0.461195
$\hat{D}_{10}$	0	1	0.596987

Note: LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$).

Table 22. LOPCOW intermediate quantities and objective weights

Selected Criterion	Dispersion Measure s_ℓ	Quadratic Magnitude q_ℓ	Information Quantity L_ℓ	Objective Weight o_ℓ
$\hat{D}_1$	0.341609	0.648636	64.1206	0.098117
$\hat{D}_2$	0.323702	0.620945	65.1419	0.099680
$\hat{D}_3$	0.349513	0.647129	61.6004	0.094261
$\hat{D}_4$	0.305010	0.684078	80.7727	0.123598
$\hat{D}_5$	0.374597	0.625620	51.2893	0.078483
$\hat{D}_6$	0.264026	0.557680	74.7738	0.114419
$\hat{D}_7$	0.372050	0.616403	50.4874	0.077256
$\hat{D}_8$	0.331369	0.623478	63.2081	0.096721
$\hat{D}_9$	0.317027	0.559649	56.8322	0.086965
$\hat{D}_{10}$	0.281267	0.659928	85.2826	0.130500

Note: LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$).

Steps 15–17: RANCOM transformation

The LOPCOW objective weights were transferred directly to the RANCOM comparison basis using Eq. (35). The corresponding values are presented in Table 23.

The RANCOM pairwise comparison matrix was constructed using Eq. (36), and the corresponding row totals were calculated using Eq. (37). The results are reported in Table 24.

Normalizing the row totals using Eq. (38) yielded the RANCOM weights presented in Table 25.

These two dependent weight representations are subsequently combined, with the RANCOM component treated as a rank-based transformation of the LOPCOW ordering rather than as an independent source of preference information.

Step 18: Combined criterion weights

The baseline calculation used $\alpha = 0.50$, giving equal numerical influence to the LOPCOW and RANCOM vectors. The resulting final criterion weights are reported in Table 26 and illustrated in Figure 3. Figure 4 compares the LOPCOW objective weights, RANCOM weights, and final combined weights across the ten retained criteria.

The largest combined weight was assigned to privacy, surveillance, and data-misuse risk ($\hat{D}_{10}$; $w_{10}^* = 0.160250$), followed by digital inclusion ($\hat{D}_4$; $w_4^* = 0.146799$) and algorithmic fairness ($\hat{D}_6$; $w_6^* = 0.132209$). Because

RANCOM was derived from the LOPCOW ordering, the two components should not be interpreted as independent weighting sources.

Table 23. LOPCOW objective weights transferred to the RANCOM comparison

Selected Criterion	LOPCOW Weight o_ℓ	RANCOM Comparison-Basis Value ϖ_ℓ
$\hat{D}_1$	0.098117	0.098117
$\hat{D}_2$	0.099680	0.099680
$\hat{D}_3$	0.094261	0.094261
$\hat{D}_4$	0.123598	0.123598
$\hat{D}_5$	0.078483	0.078483
$\hat{D}_6$	0.114419	0.114419
$\hat{D}_7$	0.077256	0.077256
$\hat{D}_8$	0.096721	0.096721
$\hat{D}_9$	0.086965	0.086965
$\hat{D}_{10}$	0.130500	0.130500

Note: LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; RANCOM = Ranking Comparison; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$).

Table 24. RANCOM comparison matrix and row totals based on LOPCOW objective weights

Criterion	$\hat{D}_1$	$\hat{D}_2$	$\hat{D}_3$	$\hat{D}_4$	$\hat{D}_5$	$\hat{D}_6$	$\hat{D}_7$	$\hat{D}_8$	$\hat{D}_9$	$\hat{D}_{10}$	t_ℓ
$\hat{D}_1$	0.5	0.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0	0.0	5.5
$\hat{D}_2$	1.0	0.5	1.0	0.0	1.0	0.0	1.0	1.0	1.0	0.0	6.5
$\hat{D}_3$	0.0	0.0	0.5	0.0	1.0	0.0	1.0	0.0	1.0	0.0	3.5
$\hat{D}_4$	1.0	1.0	1.0	0.5	1.0	1.0	1.0	1.0	1.0	0.0	8.5
$\hat{D}_5$	0.0	0.0	0.0	0.0	0.5	0.0	1.0	0.0	0.0	0.0	1.5
$\hat{D}_6$	1.0	1.0	1.0	0.0	1.0	0.5	1.0	1.0	1.0	0.0	7.5
$\hat{D}_7$	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0	0.0	0.5
$\hat{D}_8$	0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.5	1.0	0.0	4.5
$\hat{D}_9$	0.0	0.0	0.0	0.0	1.0	0.0	1.0	0.0	0.5	0.0	2.5
$\hat{D}_{10}$	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.5	9.5

Note: RANCOM = Ranking Comparison; LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$); t_ℓ = row total of the RANCOM comparison matrix. Comparison values of 1, 0.5, and 0 indicate that the row criterion has a higher, equal, or lower LOPCOW weight than the column criterion, respectively.

Table 25. RANCOM weights derived from the LOPCOW ordering

Selected Criterion	Row Total t_ℓ	RANCOM Weight b_ℓ
$\hat{D}_1$	5.5	0.11
$\hat{D}_2$	6.5	0.13
$\hat{D}_3$	3.5	0.07
$\hat{D}_4$	8.5	0.17
$\hat{D}_5$	1.5	0.03
$\hat{D}_6$	7.5	0.15
$\hat{D}_7$	0.5	0.01
$\hat{D}_8$	4.5	0.09
$\hat{D}_9$	2.5	0.05
$\hat{D}_{10}$	9.5	0.19
Total	50.0	1.00

Note: RANCOM = Ranking Comparison; LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$).

Table 26. LOPCOW, RANCOM, and final criterion weights ($\alpha = 0.50$)

Selected Criterion	LOPCOW Objective Weight o_ℓ	RANCOM Weight b_ℓ	Final Combined Criterion Weight w_ℓ^*
$\hat{D}_1$	0.098117	0.110000	0.104059
$\hat{D}_2$	0.099680	0.130000	0.114840
$\hat{D}_3$	0.094261	0.070000	0.082131
$\hat{D}_4$	0.123598	0.170000	0.146799
$\hat{D}_5$	0.078483	0.030000	0.054241
$\hat{D}_6$	0.114419	0.150000	0.132209
$\hat{D}_7$	0.077256	0.010000	0.043628
$\hat{D}_8$	0.096721	0.090000	0.093361
$\hat{D}_9$	0.086965	0.050000	0.068482
$\hat{D}_{10}$	0.130500	0.190000	0.160250

Note: LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; RANCOM = Ranking Comparison; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$); α = combination coefficient controlling the relative contributions of LOPCOW and RANCOM.

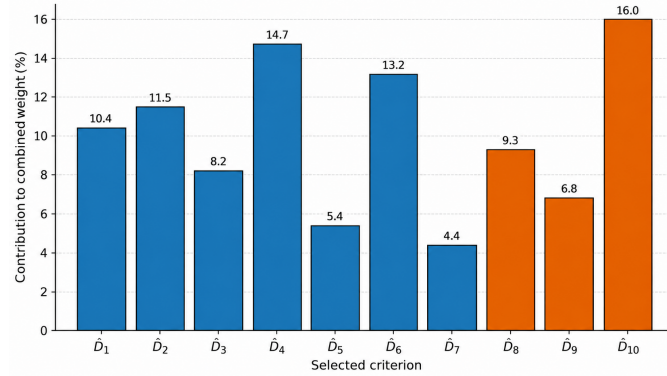


Figure 3. Final LOPCOW–RANCOM combined weights of the retained criteria

Note: LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; RANCOM = Ranking Comparison; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$). Blue bars represent benefit criteria, and orange bars represent cost criteria.

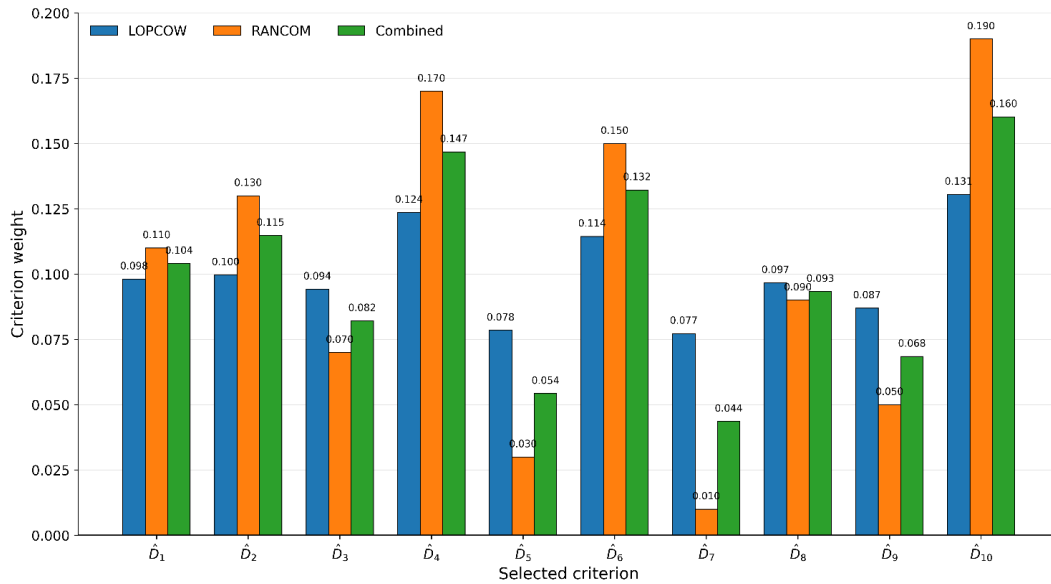


Figure 4. Comparison of LOPCOW, RANCOM, and final combined criterion weights

Note: LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; RANCOM = Ranking Comparison; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$).

Steps 19–20: MARCOS reference profiles

Eqs. (41) and (42) defined the ideal and anti-ideal reference values, which are reported in Table 27.

Table 27. Ideal and anti-ideal reference values used in MARCOS

Selected Criterion	Type	Ideal	Anti-Ideal
$\hat{D}_1$	Benefit	0.920000	0.675000
$\hat{D}_2$	Benefit	0.920000	0.765000
$\hat{D}_3$	Benefit	0.892665	0.636655
$\hat{D}_4$	Benefit	0.920000	0.475000
$\hat{D}_5$	Benefit	0.920000	0.765000
$\hat{D}_6$	Benefit	0.920000	0.704005
$\hat{D}_7$	Benefit	0.920000	0.675000
$\hat{D}_8$	Cost	0.155000	0.635000
$\hat{D}_9$	Cost	0.155000	0.635000
$\hat{D}_{10}$	Cost	0.080000	0.403358

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; $\hat{D}_\ell$ = the ℓ th retained criterion ($\ell = 1, \dots, 10$); Benefit = criterion for which a higher value is preferred; Cost = criterion for which a lower value is preferred; Ideal = best reference value for each criterion; Anti-ideal = worst reference value for each criterion.

After MARCOS normalization and application of the final criterion weights, the reference totals were $S^+ = 1.000000$ and $S^- = 0.555779$, as reported in Table 28.

Table 28. Weighted sums of the normalized MARCOS reference profiles

Reference Profile	Weighted Sum	Role
Ideal, S^+	1.000000	Upper reference
Anti-ideal, S^-	0.555779	Lower reference

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; S^+ = weighted sum of the normalized ideal reference profile; S^- = weighted sum of the normalized anti-ideal reference profile.

Step 21: Weighted MARCOS totals

Eq. (44) applied the final criterion weights to the normalized decision matrix. Summing the weighted entries produced the totals in Table 29.

G_7 obtained the largest weighted total: $S_7 = 0.880225$.

Table 29. Weighted MARCOS totals for the strategies

Strategy	Weighted Total S_i
G_1	0.756050
G_2	0.649030
G_3	0.629436
G_4	0.866791
G_5	0.822344
G_6	0.744833
G_7	0.880225
G_8	0.752988
G_9	0.602924
G_{10}	0.740737

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; G_i = the i th digital governance strategy ($i = 1, \dots, 10$); S_i = weighted sum of the normalized criterion values for the i th strategy.

Step 22: MARCOS utility degrees

Eq. (46) compared each strategy with the ideal and anti-ideal reference totals. The resulting MARCOS utility degrees are reported in Table 30.

Substituting Eq. (46) into Eq. (47) shows that the utility components f_i^- and f_i^+ are constant across all strategies.

Table 30. MARCOS utility degrees for the strategies

Strategy	K_i^-	K_i^+
G_1	1.360343	0.756050
G_2	1.167785	0.649030
G_3	1.132530	0.629436
G_4	1.559596	0.866791
G_5	1.479625	0.822344
G_6	1.340160	0.744833
G_7	1.583769	0.880225
G_8	1.354834	0.752988
G_9	1.084827	0.602924
G_{10}	1.332791	0.740737

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; G_i = the i th digital governance strategy ($i = 1, \dots, 10$); K_i^- = utility degree of the i th strategy relative to the anti-ideal reference profile; K_i^+ = utility degree of the i th strategy relative to the ideal reference profile.

Step 23: Final MARCOS appraisal scores and ranking

Eqs. (48) and (49) yielded the final MARCOS appraisal scores and strategy ranking, as reported in Table 31.

Table 31. Final MARCOS appraisal scores and strategy ranking

Rank	Symbol	Strategy	Appraisal Score	Gap From Leader
1	G_7	Mobile-first remote and island services	0.734412	0.000000
2	G_4	Citizen participation and e-consultation	0.723203	0.011209
3	G_5	Privacy-preserving urban data exchange	0.686119	0.048293
4	G_1	Unified national–municipal service portal	0.630807	0.103605
5	G_8	Climate-responsive early-warning governance	0.628253	0.106159
6	G_6	Integrated digital identity	0.621448	0.112964
7	G_{10}	Integrated intelligent governance ecosystem	0.618031	0.116381
8	G_2	AI-driven predictive service management	0.541516	0.192896
9	G_3	Urban digital-twin planning platform	0.525168	0.209244
10	G_9	Blockchain-based public administration	0.503047	0.231365

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; G_i = the i th digital governance strategy ($i = 1, \dots, 10$); appraisal score = final MARCOS score of each strategy; gap from leader = difference between the highest appraisal score and the appraisal score of the corresponding strategy; AI = artificial intelligence.

Mobile-first services for remote and island communities, G_7 , ranked first with an appraisal score of 0.734412. Citizen participation and e-consultation, G_4 , followed at 0.723203, while privacy-preserving urban data exchange, G_5 , ranked third at 0.686119. The difference between the two leading strategies was $0.734412 - 0.723203 = 0.011209$.

The baseline preference order was therefore:

$$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8 \succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9.$$

The relatively small difference between G_7 and G_4 indicates that the leading strategy should be interpreted as a narrow preference under the specified assessments and weights rather than as an overwhelmingly dominant alternative. More importantly, this ordering is conditional on the constructed decision scenario and does not establish that G_7 is universally preferable across Indonesian municipalities.

4.2 Sensitivity and Comparative Analysis

The robustness of the baseline ranking was examined from three complementary perspectives: sensitivity to the Dombi aggregation parameter κ , the assessment-profile concentration parameter τ , and the LOPCOW–RANCOM mixing parameter α ; Monte Carlo simulation under perturbed assessment conditions; and comparison with alternative weighting and ranking procedures.

Table 32 reports the MARCOS appraisal scores obtained under alternative values of the Dombi aggregation parameter κ and the assessment-profile concentration parameter τ . The tested values were selected around the baseline setting $\kappa = 2$ and $\tau = 1$, which was included explicitly to verify consistency with the main MARCOS results reported in Table 31. Across all tested parameter combinations, G_7 remained the highest-ranked strategy, while

G_4 and G_5 consistently occupied the second and third positions, respectively. The middle-ranked alternatives were comparatively more sensitive to changes in the aggregation settings, particularly at higher values of κ , whereas the leading group remained stable.

Table 32. Sensitivity of MARCOS appraisal scores and rankings to the Dombi aggregation parameter κ and assessment-profile concentration parameter τ

κ	τ	G_1	G_2	G_3	G_4	G_5	G_6	G_7	G_8	G_9	G_{10}	Resulting Ranking
1	0.5	0.6295	0.5402	0.5229	0.7207	0.6866	0.6210	0.7347	0.6280	0.5022	0.6172	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
1	1	0.6295	0.5402	0.5230	0.7205	0.6865	0.6210	0.7347	0.6281	0.5022	0.6172	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
1	2	0.6295	0.5402	0.5230	0.7204	0.6864	0.6210	0.7347	0.6281	0.5022	0.6173	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
1	5	0.6295	0.5403	0.5230	0.7203	0.6864	0.6210	0.7347	0.6281	0.5022	0.6173	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
2	0.5	0.6308	0.5415	0.5251	0.7234	0.6862	0.6215	0.7344	0.6282	0.5031	0.6180	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
2	1	0.6308	0.5415	0.5252	0.7232	0.6861	0.6214	0.7344	0.6283	0.5030	0.6180	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
2	2	0.6308	0.5415	0.5252	0.7231	0.6861	0.6214	0.7344	0.6283	0.5030	0.6181	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
2	5	0.6308	0.5416	0.5252	0.7230	0.6860	0.6214	0.7344	0.6283	0.5030	0.6181	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
3	0.5	0.6322	0.5427	0.5273	0.7266	0.6861	0.6220	0.7341	0.6286	0.5040	0.6189	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
3	1	0.6322	0.5427	0.5273	0.7264	0.6860	0.6220	0.7341	0.6286	0.5040	0.6189	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
3	2	0.6322	0.5427	0.5273	0.7263	0.6859	0.6220	0.7342	0.6286	0.5040	0.6190	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
3	5	0.6322	0.5428	0.5273	0.7262	0.6859	0.6220	0.7342	0.6287	0.5040	0.6190	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$
5	0.5	0.6374	0.5492	0.5336	0.7318	0.6831	0.6245	0.7371	0.6321	0.5057	0.6249	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_{10} \succ G_6 \succ G_2 \succ G_3 \succ G_9$
5	1	0.6375	0.5493	0.5337	0.7316	0.6830	0.6245	0.7372	0.6322	0.5056	0.6250	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_{10} \succ G_6 \succ G_2 \succ G_3 \succ G_9$
5	2	0.6375	0.5493	0.5337	0.7315	0.6829	0.6244	0.7372	0.6322	0.5056	0.6250	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_{10} \succ G_6 \succ G_2 \succ G_3 \succ G_9$
5	5	0.6375	0.5493	0.5337	0.7315	0.6829	0.6244	0.7372	0.6322	0.5056	0.6250	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_{10} \succ G_6 \succ G_2 \succ G_3 \succ G_9$

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; κ = Dombi aggregation parameter; τ = assessment-profile concentration parameter; G_i = the i th digital governance strategy ($i = 1, \dots, 10$); $\succ$ = ranks higher than.

The parameter analysis therefore indicates that the preference for G_7 is not dependent on a single choice of κ or τ , as illustrated in Figure 5. Although some middle-ranked alternatives exchanged positions under different parameter settings, the ordering of the three leading strategies remained unchanged. This provides an initial indication of ranking stability before examining the effect of the LOPCOW–RANCOM mixing parameter.

To examine whether the baseline result depended on the equal-mixing assumption between LOPCOW and RANCOM, the mixing parameter α was varied from 0 to 1. Table 33 reports the resulting MARCOS appraisal scores for $\alpha = 0.00, 0.25, 0.50, 0.75$, and 1.00, covering specifications from pure RANCOM weighting to pure LOPCOW weighting.

The sensitivity analysis with respect to the LOPCOW–RANCOM mixing parameter produced a highly stable leading group. G_7 remained the first-ranked strategy for all tested values of α , while G_4 and G_5 consistently occupied the second and third positions, respectively. As α increased from 0 to 1, the appraisal score of G_7 increased slightly, whereas that of G_4 remained almost unchanged. The gap between the two leading alternatives increased from approximately 0.0081 under the purely RANCOM-based specification to approximately 0.0142 under the purely LOPCOW-based specification. These results indicate that the first-place position of G_7 is not an artefact of the baseline choice $\alpha = 0.50$, although the magnitude of its advantage over G_4 depends modestly on the weighting specification.

Because the RANCOM weights are generated from the LOPCOW ordering, the two weighting vectors should not be interpreted as independent sources of evidence. In the present framework, the RANCOM component is instead treated as a rank-based transformation that increases the separation among criteria according to their LOPCOW-derived ordering. The sensitivity analysis with respect to α therefore evaluates whether this transformation materially changes the resulting strategy recommendation.

The parameter analysis shows that the leading group is highly stable. G_7 remained first in all 16 settings, while

G_4 and G_5 consistently occupied the second and third positions. By contrast, the relative positions of G_1 , G_6 , G_8 , and G_{10} changed under several parameter combinations. The sensitivity analysis therefore supports the robustness of the leading alternatives without implying that the complete ten-strategy ranking is invariant.

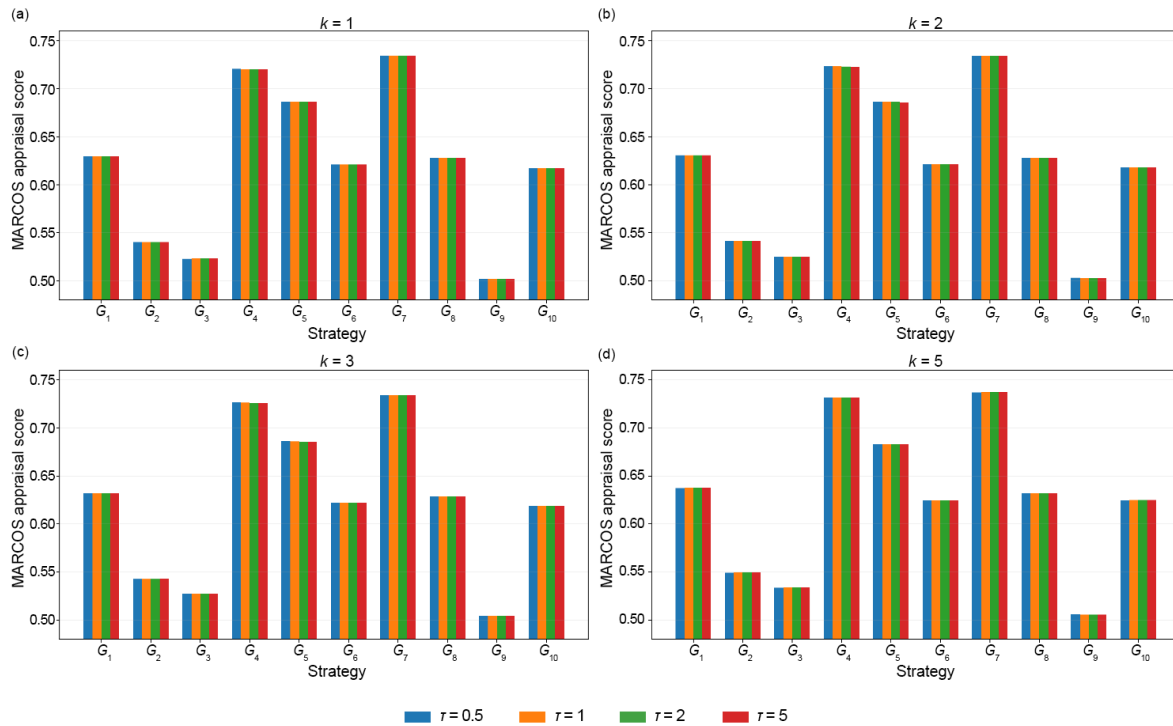


Figure 5. MARCOS appraisal scores under alternative values of the Dombi aggregation parameter κ and assessment-profile concentration parameter τ : (a) $\kappa = 1$; (b) $\kappa = 2$; (c) $\kappa = 3$; (d) $\kappa = 5$

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; G_i = the i th digital governance strategy ($i = 1, \dots, 10$).

Table 33. Sensitivity of MARCOS appraisal scores to the LOPCOW–RANCOM mixing parameter α

α	G_1	G_2	G_3	G_4	G_5	G_6	G_7	G_8	G_9	G_{10}	Gap ($A_7 - A_4$)
0.00	0.634223	0.533955	0.516937	0.723343	0.702243	0.629569	0.731484	0.624230	0.494539	0.616013	0.008141
0.25	0.632522	0.537746	0.521062	0.723295	0.694163	0.625499	0.732955	0.626249	0.498804	0.617035	0.009660
0.50	0.630821	0.541512	0.525160	0.723238	0.686115	0.621442	0.734410	0.628250	0.503042	0.618045	0.011171
0.75	0.629122	0.545252	0.529232	0.723174	0.678100	0.617399	0.735849	0.630233	0.507252	0.619043	0.012675
1.00	0.627423	0.548967	0.533276	0.723101	0.670118	0.613368	0.737272	0.632199	0.511435	0.620028	0.014171

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; RANCOM = Ranking Comparison; α = mixing parameter controlling the relative contributions of LOPCOW and RANCOM weights; G_i = the i th digital governance strategy ($i = 1, \dots, 10$); A_i = final MARCOS appraisal score of the i th strategy; $A_7 - A_4$ = difference between the appraisal scores of strategies G_7 and G_4 .

To further examine ranking robustness under uncertainty in the assessment inputs, a Monte Carlo simulation with 10,000 iterations was conducted using the ten retained criteria. The simulation was based on the criterion values obtained after criterion screening and direction restoration. In each iteration, every retained criterion value was independently perturbed using a uniform distribution within $\pm 5\%$ of that criterion's observed range. The perturbed values were constrained to remain within the corresponding baseline minimum and maximum values so that the simulated decision matrices remained within the numerical bounds of the constructed case.

After perturbation, the criteria were normalized in a common favourable direction. Criterion weights were recalculated independently in every iteration using LOPCOW, Entropy, Method based on the Removal Effects of Criteria (MEREC), and Criteria Importance Through Intercriteria Correlation (CRITIC). For each resulting weight vector, the alternatives were ranked using TOPSIS, VIKOR, Multi-Objective Optimization on the Basis of Ratio

Analysis (MOORA), Multi-Attributive Border Approximation Area Comparison (MABAC), Weighted Aggregated Sum Product Assessment (WASPAS), EDAS, and Additive Ratio Assessment (ARAS). For the MARCOS specification, the LOPCOW ordering generated the corresponding RANCOM comparison weights in each iteration, after which the two weight vectors were combined using $\alpha = 0.50$ before MARCOS ranking was performed. The first-ranked strategy was recorded for every weighting–ranking combination.

Table 34 reports the resulting first-place frequencies from the 10,000 Monte Carlo iterations.

Table 34. Dominant first-ranked strategies in 10,000 Monte Carlo simulations

Ranking Method	LOPCOW	Entropy	MEREC	CRITIC
TOPSIS	G_7 (99.11%)	G_4 (39.61%)	G_{10} (75.45%)	G_{10} (98.56%)
VIKOR	G_7 (99.47%)	G_8 (99.97%)	G_8 (100.00%)	G_8 (99.20%)
MOORA	G_7 (73.15%)	G_4 (74.18%)	G_{10} (58.54%)	G_{10} (100.00%)
MABAC	G_7 (73.76%)	G_4 (67.90%)	G_{10} (75.06%)	G_{10} (100.00%)
WASPAS	G_7 (99.93%)	G_8 (89.60%)	G_8 (92.45%)	G_8 (100.00%)
EDAS	G_4 (54.96%)	G_4 (65.24%)	G_{10} (68.47%)	G_{10} (100.00%)
ARAS	G_7 (66.49%)	G_4 (65.32%)	G_{10} (67.90%)	G_{10} (100.00%)
MARCOS	Hybrid: G_7 (72.75%)	—	—	—

Note: LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; MEREC = Method based on the Removal Effects of Criteria; CRITIC = Criteria Importance Through Intercriteria Correlation; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; VIKOR = multi-criteria optimization and compromise solution; MOORA = Multi-Objective Optimization on the Basis of Ratio Analysis; MABAC = Multi-Attributive Border Approximation Area Comparison; WASPAS = Weighted Aggregated Sum Product Assessment; EDAS = Evaluation based on Distance from Average Solution; ARAS = Additive Ratio Assessment; MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; RANCOM = Ranking Comparison; G_i = the i th digital governance strategy ($i = 1, \dots, 10$); Hybrid = MARCOS using combined LOPCOW–RANCOM weights. Percentages indicate the proportion of the 10,000 Monte Carlo simulations in which the corresponding strategy ranked first; “—” = results not reported.

Complete first-place counts for all ten strategies under each weighting–ranking combination are provided in Table A1.

The Monte Carlo results show that ranking robustness depends materially on both the weighting and ranking procedures. Under LOPCOW weighting, G_7 remained the dominant first-ranked strategy for TOPSIS, VIKOR, MOORA, MABAC, WASPAS, and ARAS, with first-place frequencies ranging from 66.49% to 99.93%, as shown in Figure 6a. EDAS with LOPCOW showed a closer competition between G_4 and G_7 , with G_4 ranking first in 54.96% of the simulations and G_7 in 45.04%.

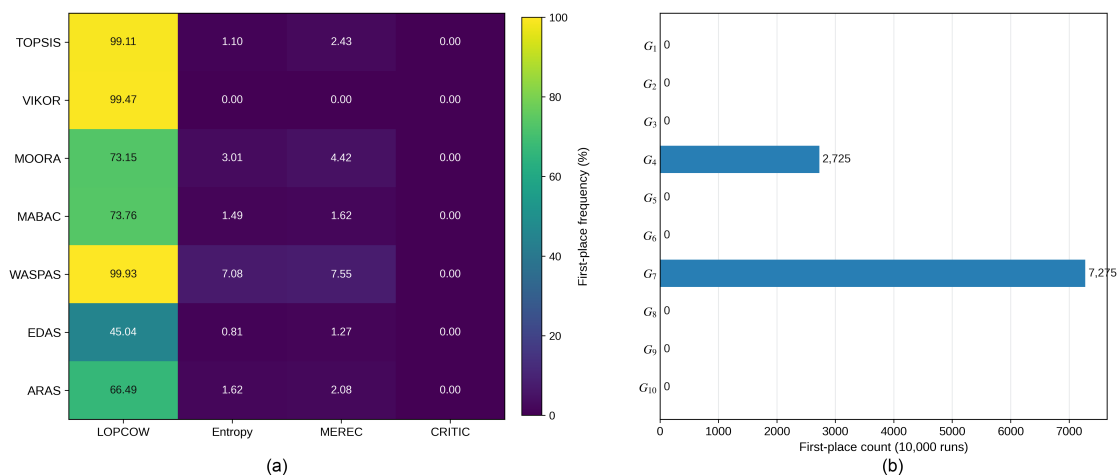


Figure 6. First-place outcomes from 10,000 Monte Carlo simulations: (a) first-place frequency of G_7 under alternative weighting–ranking specifications; (b) first-place counts under the hybrid-weighted MARCOS framework

Note: TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; VIKOR = Multicriteria Optimization and Compromise Solution; MOORA = Multi-Objective Optimization by Ratio Analysis; MABAC = Multi-Attributive Border Approximation Area Comparison; WASPAS =

Weighted Aggregated Sum Product Assessment; EDAS = Evaluation based on Distance from Average Solution; ARAS = Additive Ratio Assessment; LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; MEREC = Method based on the Removal Effects of Criteria; CRITIC = Criteria Importance Through Intercriteria Correlation; MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; G_i = the i th digital governance strategy ($i = 1, \dots, 10$).

Greater variation emerged when Entropy, MEREC, and CRITIC weights were used. Depending on the ranking method, G_4 , G_8 , or G_{10} became the most frequent first-ranked strategy. This result indicates that the identity of the leading alternative is more sensitive to the weighting specification than suggested by the parameter analysis alone.

Under the hybrid-weighted MARCOS specification, G_7 ranked first in 7,275 of the 10,000 simulations, corresponding to 72.75%, while G_4 ranked first in the remaining 2,725 simulations, as shown in Figure 6b. G_7 therefore remained the dominant alternative under the proposed MARCOS framework, although the simulation confirms that its advantage is not invariant to perturbations in the assessment inputs.

These frequencies quantify ranking stability within the specified perturbation design and should not be interpreted as probabilities of successful real-world implementation. Differences among the comparator configurations reflect changes in both criterion weighting and ranking procedures.

Table 35 compares the rankings obtained using additional MCDM procedures. For comparability, the alternative MCDM procedures were applied to the present decision problem; the cited studies are provided as methodological references and did not evaluate the G_1 – G_{10} strategies themselves.

Table 35. Comparison of rankings obtained by applying alternative MCDM methods to the present decision problem

Methodological Reference	MCDM Method	Ranking	First-Ranked Strategy
Singh and Kumar [31]	AHP	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_2 \succ G_{10} \succ G_3 \succ G_9$	G_7
Bardis [33]	PROMETHEE	$G_7 \succ G_4 \succ G_5 \succ G_8 \succ G_1$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$	G_7
Ayyildiz et al. [27]	SWARA–TOPSIS	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_3 \succ G_2 \succ G_9$	G_7
Janani et al. [17]	Interval-valued intuitionistic fuzzy CoCoSo	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_{10} \succ G_6 \succ G_2 \succ G_3 \succ G_9$	G_7
Mishra et al. [24]	Picture fuzzy modified CoCoSo	$G_7 \succ G_4 \succ G_5 \succ G_8 \succ G_1$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_9 \succ G_3$	G_7
Augustin [28]	Picture fuzzy EDAS–BW	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_2 \succ G_{10} \succ G_9 \succ G_3$	G_7
Şahan et al. [40]	Fuzzy DEMATEL–COPRAS	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_6$ $\succ G_8 \succ G_2 \succ G_{10} \succ G_3 \succ G_9$	G_7
Present study	MARCOS	$G_7 \succ G_4 \succ G_5 \succ G_1 \succ G_8$ $\succ G_6 \succ G_{10} \succ G_2 \succ G_3 \succ G_9$	G_7

Note: MCDM = multi-criteria decision-making; AHP = Analytic Hierarchy Process; PROMETHEE = Preference Ranking Organization Method for Enrichment Evaluations; SWARA = Step-wise Weight Assessment Ratio Analysis; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; CoCoSo = Combined Compromise Solution; EDAS = Evaluation based on Distance from Average Solution; BW = best–worst method; DEMATEL = Decision-Making Trial and Evaluation Laboratory; COPRAS = Complex Proportional Assessment; MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; G_i = the i th digital governance strategy ($i = 1, \dots, 10$); $\succ$ = ranks higher than.

All comparison procedures identified G_7 as the leading strategy, while G_4 and G_5 consistently occupied the next two positions. Differences among methods were concentrated mainly in the middle and lower parts of the ranking.

Across all tested parameter combinations, G_7 remained the highest-ranked strategy, while G_4 and G_5 retained the second and third positions, respectively. The appraisal scores varied only moderately as κ and τ changed, indicating that the leading recommendation was not driven by a single aggregation setting. Some middle-ranked alternatives exchanged positions, particularly under higher values of κ , showing that the complete ranking is more sensitive than the leading group. Taken together with the α -sensitivity analysis and the Monte Carlo simulation, these results indicate that the main ranking conclusion is comparatively robust within the constructed decision setting, while lower-order positions remain dependent on modelling assumptions.

5 Discussion

Mobile-first services for remote and island communities G_7 achieved the highest MARCOS appraisal score, 0.734412, followed by citizen participation and e-consultation G_4 at 0.723203 and privacy-preserving urban data exchange G_5 at 0.686119. The difference between G_7 and G_4 was only 0.011209, indicating that the baseline model identified a preferred alternative but did not produce a wide separation between the two leading strategies. The complete ranking is reported in Table 31.

The first-place result for G_7 is meaningful only within the structure of the present case study. The assessment of this strategy reflected the importance of service accessibility, digital inclusion, implementation burden, and privacy-related risk among the retained criteria. Its leading position therefore suggests that, under the specified

assessments, reducing geographic and procedural barriers to public services provides a strong overall balance across the evaluated dimensions. This is particularly relevant in the Indonesian context, where differences in connectivity, physical access to government offices, and the availability of assisted service channels may substantially influence the practical value of digital public services.

The appraisal score does not, however, indicate how many residents would actually adopt a mobile channel, whether they would be able to complete transactions independently, or how much reliable service provision would cost in a particular locality. These questions require locally observed evidence. The result should therefore be interpreted as a structured comparison of strategic options rather than as an empirical estimate of service uptake or implementation success.

The near-tie between G_7 and G_4 is substantively important. The two alternatives address different dimensions of urban governance. G_7 focuses primarily on how residents obtain access to public services, whereas G_4 concerns how residents participate in public decision-making, communicate local concerns, and follow government responses. If a municipality faces severe barriers to routine service access, a mobile-first strategy may deserve greater consideration. Conversely, where the more pressing issue is limited participation in local planning and service development, a citizen-participation and e-consultation system may be more appropriate. The current assessments do not establish which of these needs is more urgent in any specific municipality. The small numerical gap therefore reinforces the importance of defining the local governance problem before interpreting the ranking as a basis for action.

G_5 , the privacy-preserving urban data-exchange platform, ranked third and remained relatively close to the two leading strategies. Its result highlights the role of inter-agency data coordination in digital urban governance. Improved data exchange can reduce repeated information requests, support cross-agency referrals, and improve administrative continuity, but these advantages depend on clear access rights, data-quality standards, retention policies, and institutional accountability. The difference between G_5 and G_7 was 0.048293, which is larger than the gap between the first two alternatives but still does not indicate an overwhelming separation.

Several strategies in the middle of the ranking were also separated by relatively small margins. The unified national-municipal service portal G_1 , with an appraisal score of 0.630807, exceeded climate-responsive early-warning governance G_8 , at 0.628253, by only 0.002554. Integrated digital identity G_6 , at 0.621448, and the integrated intelligent governance ecosystem G_{10} , at 0.618031, were similarly close. These numerical differences should be reported accurately, but their decimal precision should not be interpreted as evidence of substantial practical differences in policy value. Each alternative addresses a different urban-service problem, and the choice among them ultimately depends on the policy objective, local conditions, resource constraints, and implementation capacity of the authority concerned.

AI-driven predictive service management G_2 , the urban digital-twin planning platform G_3 , and blockchain-based public administration G_9 occupied the final three positions, with appraisal scores of 0.541516, 0.525168, and 0.503047, respectively. These positions should not be interpreted as evidence that predictive analytics, digital twins, or blockchain are inherently unsuitable for urban governance. The alternatives were evaluated as broad strategic programmes under a common set of criteria. A narrowly defined predictive model for one municipal service, a task-specific urban digital twin, or an auditable workflow system may have substantially different costs, data requirements, institutional demands, and expected benefits. Such applications would require separate assessment under criteria appropriate to their specific purpose.

The criterion-screening stage provides further insight into the resulting ranking. MI retained ten of the fifteen original criteria. The retained benefit criteria were public-service accessibility, service-delivery efficiency, psychological acceptance, digital inclusion, transparency and accountability, algorithmic fairness, and climate and environmental resilience. The retained cost criteria were initial implementation cost, data-integration cost, and privacy, surveillance, and data-misuse risk. Citizen trust, operation and maintenance cost, implementation time, technological and administrative complexity, and vendor dependence were removed under the stated redundancy rule.

Removal from the final weighted matrix does not imply that these concerns cease to matter in practice. A municipality would still need to finance system maintenance, assess implementation time, manage technological and administrative complexity, and evaluate its dependence on external vendors. The screening procedure only indicates that, within the constructed decision matrix, these criteria contributed information that overlapped sufficiently with other retained criteria to trigger the specified redundancy rule.

Among the retained criteria, privacy, surveillance, and data-misuse risk received the largest final weight: $w_{10}^* = 0.160250$. Digital inclusion received a weight of $w_4^* = 0.146799$, and algorithmic fairness received $w_6^* = 0.132209$. By contrast, climate and environmental resilience received the smallest final weight: $w_7^* = 0.043628$. These values are properties of the constructed assessment matrix and the specified weighting procedure. They should not be interpreted as estimates of the priorities of Indonesian residents, municipal officials, or public institutions. In particular, the RANCOM comparisons were generated from the LOPCOW ordering. Combining the two vectors with $\alpha = 0.50$ therefore does not create an independent expert-preference component.

The MARCOS reference profiles provide another perspective on the final result. The weighted total for G_7

was $S_7 = 0.880225$ compared with $S_4 = 0.866791$ for G_4 . The final appraisal calculation preserved this order. These values are useful for comparing alternatives within the model, but they are not probabilities of successful implementation, predicted adoption rates, or measures of citizen support.

The sensitivity and comparative analyses provide a more nuanced view of ranking robustness. Across the tested combinations of the Dombi aggregation parameter and the assessment-profile concentration parameter, G_7 remained first, while G_4 and G_5 consistently occupied the second and third positions. The additional sensitivity analysis of the LOPCOW–RANCOM mixing parameter produced the same leading order across the full range from $\alpha = 0$ to $\alpha = 1$. The advantage of G_7 over G_4 increased gradually as greater weight was assigned to the LOPCOW component, but the identity of the first-ranked strategy did not change. This finding shows that the baseline result is not driven solely by the equal-mixing assumption at $\alpha = 0.50$.

These findings indicate that the baseline preference for G_7 is robust within the proposed framework but is not invariant to alternative modelling choices. The result is consistent with the small baseline difference between G_7 and G_4 and reinforces the need to interpret the ranking as conditional on the assessment inputs, criterion-weighting method, and ranking rule.

The principal limitation of the study is therefore not computational instability but the source of the input information. The assessment matrices and role-based assessment profiles were constructed to demonstrate the decision-support procedure; they were not derived from direct observations of implemented strategies, surveys of residents, or elicited judgments from municipal officials. Consequently, the analysis supports a methodological and comparative conclusion rather than a general empirical claim about digital-governance priorities across Indonesia.

From an urban-management perspective, the main value of the framework lies in making the basis of a strategic choice explicit. A municipal authority can identify the alternatives under consideration, define the relevant benefit and cost criteria, examine overlap among those criteria, document the origin of criterion weights, and assess whether a preferred option remains stable when assumptions change. This traceability is especially important in AI-enabled public-service transformation because the apparent technical attractiveness of a strategy may conflict with requirements for inclusion, accountability, privacy, institutional capacity, and long-term implementation feasibility.

For practical use, the constructed inputs should therefore be replaced with locally observed service-delivery data, realistic implementation and operating costs, and assessments obtained from residents, public officials, and responsible agencies. Local application may also alter the criterion set itself. The most defensible conclusion from the present numerical case is consequently that G_7 is the highest-ranked strategy under the stated assessments and modelling assumptions, with G_4 remaining a close second. The framework is intended to support transparent comparison of such alternatives rather than prescribe a universal digital-governance strategy for Indonesian cities.

6 Conclusion and Future Directions

This study developed an integrated decision-support framework for comparing alternative digital urban governance and public-service transformation strategies in Indonesia. PiFNs were used to preserve positive, neutral, and negative components of the initial assessments. MI was then applied to identify overlapping criteria, LOPCOW derived objective criterion weights from the reduced decision matrix, RANCOM transformed the LOPCOW ordering into comparison-based weights, and MARCOS produced the final strategy ranking.

The screening procedure retained ten of the fifteen original criteria. Under the baseline configuration, mobile-first services for remote and island communities G_7 ranked first with an appraisal score of 0.734412. Citizen participation and e-consultation G_4 ranked second at 0.723203, followed by privacy-preserving urban data exchange G_5 at 0.686119. The difference between G_7 and G_4 was only 0.011209, indicating that the leading alternative was preferred under the specified assessments but did not dominate the second-ranked strategy by a large margin.

The robustness analyses showed that G_7 remained first across all tested combinations of the Dombi aggregation parameter and the assessment-profile concentration parameter. In the Monte Carlo analysis, G_7 also remained the dominant first-ranked strategy under the proposed hybrid-weighted MARCOS framework, appearing first in 72.75% of the 10,000 simulations. However, alternative weighting and ranking specifications produced greater variation, with G_4 , G_8 , or G_{10} becoming the most frequent first-ranked strategy in several comparator configurations. The results therefore support the stability of the baseline recommendation within the proposed framework while also demonstrating that the final ranking remains sensitive to modelling choices.

The methodological contribution of the study lies in the traceable separation of assessment representation, criterion screening, weighting, and strategy ranking. This structure makes it possible to identify where uncertainty enters the decision process, which criteria are removed as redundant, how criterion weights are generated, and how sensitive the resulting preference order is to alternative assumptions. The framework therefore provides a transparent basis for urban authorities seeking to compare digital-governance strategies that involve competing objectives related to access, inclusion, institutional feasibility, privacy, fairness, environmental resilience, and implementation burden.

The study also has important limitations. Most importantly, the assessment matrices and role-based assessment profiles were constructed for methodological demonstration rather than collected from residents, municipal officials,

or operating public services. In addition, the RANCOM comparison values were derived from the LOPCOW ordering and therefore do not constitute an independent source of expert preference. These limitations constrain the interpretation of the results to the specified numerical scenario.

Future research should apply the framework to actual municipalities using observed service-delivery data, verified implementation and operating costs, and assessments elicited from the agencies and communities affected by the proposed services. Particular attention should be given to residents who experience difficulty accessing existing digital channels, including those facing geographic, socioeconomic, disability-related, or digital-literacy barriers.

A further extension should obtain criterion preferences independently from experts, public officials, and citizens rather than derive them from the objective-weight ordering. This would allow a clearer comparison between data-driven criterion importance and stakeholder priorities. The Monte Carlo perturbations were applied independently to the retained criterion values and therefore did not explicitly preserve potential dependence structures among the retained criteria. Future applications could incorporate correlated perturbations or empirically estimated joint distributions when sufficient observational data are available. Future studies should also examine the sensitivity of the results to alternative criterion sets, screening thresholds, assessment scales, and aggregation procedures.

The close scores obtained for G_7 and G_4 indicate that future empirical applications should pay particular attention to the trade-off between service accessibility and citizen participation. In practice, these strategies need not always be mutually exclusive. Municipalities may ultimately adopt portfolios or staged combinations of digital-governance interventions rather than select a single isolated strategy.

Overall, the proposed framework demonstrates how competing digital urban governance strategies can be compared in a transparent and reproducible manner while preserving the distinction between model-based ranking and real-world policy evidence. Its practical value lies not in identifying one universally optimal strategy, but in providing urban authorities with a structured approach for documenting assumptions, examining trade-offs, testing ranking stability, and supporting accountable decisions on future public-service transformation.

Author Contributions

Conceptualization, M.F.Z., S.A. and G.A.; methodology, M.F.Z. and G.A.; software, M.F.Z.; validation, M.F.Z., S.A. and G.A.; formal analysis, M.F.Z. and G.A.; investigation, M.F.Z. and S.A.; resources, S.A.; data curation, M.F.Z. and G.A.; writing—original draft preparation, M.F.Z. and G.A.; writing—review and editing, M.F.Z., S.A., and G.A.; visualization, M.F.Z., S.A., and G.A.; supervision, S.A.; project administration, S.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] D. E. Luna, S. Picazo-Vela, B. Buyannemekh, and L. F. Luna-Reyes, “Creating public value through digital service delivery from a citizen’s perspective,” *Gov. Inf. Q.*, vol. 41, no. 2, p. 101928, 2024. <https://doi.org/10.1016/j.giq.2024.101928>
- [2] S. Dechamps, A. Simonofski, and C. Burnay, “Citizen-centricity in digital government: A theoretical and empirical typology,” *Gov. Inf. Q.*, vol. 42, no. 1, p. 102005, 2025. <https://doi.org/10.1016/j.giq.2024.102005>
- [3] M. Al-Kautsar Maktub, P. W. Handayani, and F. P. Sunarso, “Citizen acceptance and use of the Jakarta Kini (JAKI) e-government: Extended unified model for electronic government adoption,” *Heliyon*, vol. 11, no. 2, p. e42078, 2025. <https://doi.org/10.1016/j.heliyon.2025.e42078>
- [4] D. Varastika, M. E. Wijaya, M. S. C. Aurelie, P. W. Handayani, and H. Fitriani, “Mobile passport success in Indonesia based on the DeLone–McLean information systems success model,” *Telemat. Inform. Rep.*, vol. 20, p. 100251, 2025. <https://doi.org/10.1016/j.teler.2025.100251>
- [5] H. Liu, Q. Zhou, and S. Liang, “Digital inclusion in public services for vulnerable groups: A systematic review for research themes and goal-action framework from the lens of public service ecosystem theory,” *Gov. Inf. Q.*, vol. 42, no. 2, p. 102019, 2025. <https://doi.org/10.1016/j.giq.2025.102019>
- [6] G. H. Djatmiko, O. Sinaga, and S. Pawirosumarto, “Digital transformation and social inclusion in public services: A qualitative analysis of e-government adoption for marginalized communities in sustainable governance,” *Sustainability*, vol. 17, no. 7, p. 2908, 2025. <https://doi.org/10.3390/su17072908>
- [7] R. Purnamasari, A. I. Hasanudin, R. Zulfikar, and H. Yazid, “Technological infrastructure and financial resource availability in enhancing public services and government performance: The role of digital innovation adoption in Indonesia,” *Soc. Sci. Humanit. Open*, vol. 11, p. 101621, 2025. <https://doi.org/10.1016/j.ssaho.2025.101621>

- [8] K. Tissayakorn, "Comparative analysis of e-government development status of ASEAN member states: Accomplishments and challenges," *Sustain. Futures*, vol. 10, p. 101353, 2025. <https://doi.org/10.1016/j.sfr.2025.101353>
- [9] O. Agbabiaka, A. Ojo, and N. Connolly, "Requirements for trustworthy AI-enabled automated decision-making in the public sector: A systematic review," *Technol. Forecast. Soc. Change*, vol. 215, p. 124076, 2025. <https://doi.org/10.1016/j.techfore.2025.124076>
- [10] P. G. R. de Almeida and C. D. dos Santos Júnior, "Artificial intelligence governance: Understanding how public organizations implement it," *Gov. Inf. Q.*, vol. 42, no. 1, p. 102003, 2025. <https://doi.org/10.1016/j.giq.2024.102003>
- [11] D. Catalá-Pérez, M. de-Miguel-Molina, B. de-Miguel-Molina, and B. García-Ortega, "Unveiling digital public services: A citizen-led hybrid approach with mystery shoppers and fsQCA," *Telemat. Inform. Rep.*, vol. 20, p. 100257, 2025. <https://doi.org/10.1016/j.teler.2025.100257>
- [12] S. Halim and B. Bounabat, "From assessment to action: A decision-support methodology for digital government transformation," *Sustainability*, vol. 18, no. 9, p. 4362, 2026. <https://doi.org/10.3390/su18094362>
- [13] L. A. Zadeh, "Fuzzy sets," *Inf. Control*, vol. 8, no. 3, pp. 338–353, 1965. [https://doi.org/10.1016/S0019-9958\(65\)90241-X](https://doi.org/10.1016/S0019-9958(65)90241-X)
- [14] F. N. Shahmohammad, Y. Pourrahimian, and N. Akbari-Gharalari, "Synthesizing complexity: Trends, challenges, and future directions in fuzzy-based multicriteria decision-making (FMCDM) methods," *Appl. Soft Comput.*, vol. 167, p. 112362, 2024. <https://doi.org/10.1016/j.asoc.2024.112362>
- [15] K. T. Atanassov, "Intuitionistic fuzzy sets," *Fuzzy Sets Syst.*, vol. 20, no. 1, pp. 87–96, 1986. [https://doi.org/10.1016/S0165-0114\(86\)80034-3](https://doi.org/10.1016/S0165-0114(86)80034-3)
- [16] M. Ortiz-Barrios, N. Jaramillo-Rueda, A. Espeleta-Aris, B. Kucukaltan, and L. Cuenca, "Integrated fuzzy decision-making methodology with intuitionistic fuzzy numbers: An application for disaster preparedness in clinical laboratories," *Expert Syst. Appl.*, vol. 263, p. 125712, 2025. <https://doi.org/10.1016/j.eswa.2024.125712>
- [17] K. Janani, S. S. Mohanrasu, A. Kashkynbayev, and R. Rakkiyappan, "Ensemble feature selection via CoCoSo method extended to interval-valued intuitionistic fuzzy environment," *Math. Comput. Simul.*, vol. 229, pp. 50–77, 2025. <https://doi.org/10.1016/j.matcom.2024.09.023>
- [18] Y. Liu, D. Tang, Z. Huang, and Z. Wang, "A novel knowledge-driven intuitionistic fuzzy multi-criteria decision-making method for evaluating the operating condition of the aluminum electrolysis cell," *J. Taiwan Inst. Chem. Eng.*, vol. 175, p. 106256, 2025. <https://doi.org/10.1016/j.jtice.2025.106256>
- [19] F. Gao, "Evaluation of sustainable airport development strategies using an integrated intuitionistic fuzzy decision-making method and unknown weights," *Alex. Eng. J.*, vol. 127, pp. 1109–1125, 2025. <https://doi.org/10.1016/j.aej.2025.07.001>
- [20] M. M. I. Mahin, M. J. B. Rouf, S. Salman, S. M. Morshed, and M. Morshed, "An intuitionistic fuzzy analytics approach for examining factors in decentralized renewable energy adoption in emerging economies," *Decis. Anal. J.*, vol. 18, p. 100662, 2026. <https://doi.org/10.1016/j.dajour.2025.100662>
- [21] B. C. Cuong, "Picture fuzzy sets," *J. Comput. Sci. Cybern.*, vol. 30, no. 4, pp. 409–420, 2014. <https://doi.org/10.15625/1813-9663/30/4/5032>
- [22] T. Punetha, Komal, and D. Pamucar, "Novel picture fuzzy power partitioned Hamy mean operators with Dempster–Shafer theory and their applications in MCDM," *Artif. Intell. Rev.*, vol. 57, no. 6, p. 145, 2024. <https://doi.org/10.1007/s10462-024-10757-8>
- [23] K. Shi and J. Peng, "An online review-driven picture fuzzy multi-criteria group decision-making approach for evaluating the online medical service quality of doctors," *Symmetry*, vol. 16, no. 6, p. 639, 2024. <https://doi.org/10.3390/sym16060639>
- [24] A. R. Mishra, S. M. Chen, and P. Rani, "Multi-attribute decision-making based on picture fuzzy distance measure-based relative closeness coefficients and modified combined compromise solution method," *Inf. Sci.*, vol. 664, p. 120325, 2024. <https://doi.org/10.1016/j.ins.2024.120325>
- [25] S. Ashraf, C. Jana, M. Sohail, R. Choudhary, S. Ahmad, and M. Deveci, "Multi-criteria decision-making model based on picture hesitant fuzzy soft set approach: An application of sustainable solar energy management," *Inf. Sci.*, vol. 686, p. 121334, 2025. <https://doi.org/10.1016/j.ins.2024.121334>
- [26] L. Ocampo, D. P. Abellana, J. Selle, R. Namoco, F. Maturan, J. K. Padro, and S. S. Evangelista, "Dynamical system modelling for picture fuzzy multi-criteria public opinion on transportation infrastructure," *Math. Comput. Simul.*, vol. 236, pp. 455–474, 2025. <https://doi.org/10.1016/j.matcom.2025.04.024>
- [27] E. Ayyildiz, T. K. Karaca, B. Kara, B. Y. Kavus, and N. Aydin, "Picture-fuzzy decision-making tool for enhanced risk prioritization in construction and demolition waste management: A hybrid FMEA–Fine–Kinney–SWARA–TOPSIS approach," *Buildings*, vol. 15, no. 22, p. 4143, 2025. <https://doi.org/10.3390/buildings15224143>
- [28] F. Augustin, "Enhanced picture fuzzy and centrality based EDAS-BW approach and its application in healthcare,"

Results Eng., vol. 26, p. 105273, 2025. <https://doi.org/10.1016/j.rineng.2025.105273>

- [29] G. Sirbiladze, H. Garg, B. Midodashvili, I. Parshutkin, and B. Ghvaberidze, "Decision-making framework with q-rung picture fuzzy linguistic information and its applications to logistics hubs during disaster response establishment," *Results Control Optim.*, vol. 19, p. 100541, 2025. <https://doi.org/10.1016/j.rico.2025.100541>
- [30] U. Ali, J. Li, T. Senapati, S. A. Edalatpanah, S. Duleba, and S. Moslem, "A hybrid decision-making model in a complex Fermatean picture fuzzy environment with application to emergency management," *Complex Intell. Syst.*, vol. 12, no. 1, p. 15, 2026. <https://doi.org/10.1007/s40747-025-02112-3>
- [31] S. Singh and V. Kumar, "Modelling the determinants for sustainable smart city through interpretive structure modelling and analytic hierarchy process," *Comput. Urban Sci.*, vol. 4, no. 1, p. 16, 2024. <https://doi.org/10.1007/s43762-024-00125-1>
- [32] M. Shulajkovska, M. Smerkol, G. Noveski, M. Bohanec, and M. Gams, "Artificial intelligence-based decision support system for sustainable urban mobility," *Electronics*, vol. 13, no. 18, p. 3655, 2024. <https://doi.org/10.3390/electronics13183655>
- [33] G. Bardis, "A declarative modeling framework for intuitive multiple criteria decision analysis in a visual semantic urban planning environment," *Electronics*, vol. 13, no. 23, p. 4845, 2024. <https://doi.org/10.3390/electronics13234845>
- [34] A. P. Rachman, C. Ichwania, R. A. Mangkuto, J. Pradipta, M. D. Koerniawan, and J. Sarwono, "Comparison of multi-criteria decision-making methods for selection of optimum passive design strategy," *Energy Build.*, vol. 314, p. 114285, 2024. <https://doi.org/10.1016/j.enbuild.2024.114285>
- [35] M. A. Mujtaba, A. Munir, S. Imran, M. K. Nasir, M. G. Muhayyuddin, A. Javed, A. Mehmood, M. A. Habila, H. Fayaz, and A. Qazi, "Evaluating sustainable municipal solid waste management scenarios: A multicriteria decision making approach," *Heliyon*, vol. 10, no. 4, p. e25788, 2024. <https://doi.org/10.1016/j.heliyon.2024.e25788>
- [36] G. Kou, H. Dinçer, S. Yüksel, and F. S. Alotaibi, "A neuro decision-making approach for prioritizing circular economy criteria in sustainable smart cities," *Heliyon*, vol. 10, no. 22, p. e40354, 2024. <https://doi.org/10.1016/j.heliyon.2024.e40354>
- [37] M. R. Valipour Malakshah and Z. Amini, "Intersection importance assessment for an operationally resilient urban traffic network: A multi-criteria decision-making-based framework," *Reliab. Eng. Syst. Saf.*, vol. 266, p. 111661, 2025. <https://doi.org/10.1016/j.res.2025.111661>
- [38] S. Subash and N. Narmadhai, "Optimal location for electric vehicle charging station using multiobjective assessment method," *Sustain. Energy Res.*, vol. 12, no. 1, p. 36, 2025. <https://doi.org/10.1186/s40807-025-00184-w>
- [39] Y. H. Lin, M. Qiao, and Z. E. Nwetlawung, "A DANP model for evaluating the factors influencing blockchain adoption in the construction industry's carbon trading," *Urban Lifeline*, vol. 3, no. 1, p. 10, 2025. <https://doi.org/10.1007/s44285-025-00045-4>
- [40] M. Şahan, Z. S. Kara, and I. Yilmaz, "Selection of marketing strategies in the retail industry through integrated fuzzy DEMATEL and COPRAS methods," *Int. J. Fuzzy Syst.*, vol. 28, no. 1, pp. 36–51, 2026. <https://doi.org/10.1007/s40815-025-01977-3>
- [41] R. Ventura, M. Carra, and B. Barabino, "Citizen-based decision support for sustainable public transport system selection: Evidence from Vietnam," *J. Urban Mobil.*, vol. 10, p. 100247, 2026. <https://doi.org/10.1016/j.urbmob.2026.100247>
- [42] D. Triantafyllidou and D. G. Vagiona, "Enhancing wind farm siting with the combined use of multi-criteria decision-making methods," *Wind*, vol. 6, no. 1, p. 4, 2026. <https://doi.org/10.3390/wind6010004>
- [43] U. Pribadi, M. Iqbal, M. A. Ibrahim, Juhari, and Y. Ahdarrijal, "Nexus of public organization, transparency, and accountability in Indonesia's digital governance," *J. Public Aff.*, vol. 24, no. 3, p. e2940, 2024. <https://doi.org/10.1002/pa.2940>
- [44] R. Rahmatullah, A. Habibi, K. Khaeruddin, L. N. Yaqin, T. M. Alharmali, M. S. O. Fauzee, and J. Mahat, "A study of user satisfaction and net benefits in Indonesia through the DeLone and McLean model for e-government success," *Discov. Sustain.*, vol. 6, no. 1, p. 710, 2025. <https://doi.org/10.1007/s43621-025-01645-4>
- [45] A. Pratiwi, M. E. Rahmawyanet, P. A. Wibowo Putra, and D. I. Sensuse, "Systematic literature review on artificial intelligence in Indonesia's public sector: Reimagining digital government," *JITK (J. Ilmu Pengetah. Teknol. Komput.)*, vol. 11, no. 2, pp. 304–316, 2025. <https://doi.org/10.33480/jitk.v11i2.6842>
- [46] L. Fitriani, M. L. Khodra, and K. Surendro, "A conceptual framework for AI adoption in business architecture with case studies in higher education and government," *Discov. Artif. Intell.*, vol. 5, no. 1, p. 409, 2025. <https://doi.org/10.1007/s44163-025-00673-3>
- [47] Safaruddin, I. A. M. Ishak, and A. Ariyadi, "AI-driven governance and the transformation of central–local authority in Indonesia," *Gov. Resil.*, vol. 4, no. 1, pp. 1–21, 2026. <https://doi.org/10.62503/gr.v4i1.46>

- [48] A. Widuri and A. Jarkasih, “The influence of artificial intelligence readiness on digital public service innovation in Indonesian local governments,” *Contemp. Public Adm. Rev.*, vol. 4, no. 1, pp. 87–114, 2026. <https://doi.org/10.26593/copar.v4i1.10689>
- [49] Y. Nugroho and S. Antari, “Rethinking the governance of artificial intelligence oversight in Indonesia from a global perspective,” *Mendapo J. Adm. Law*, vol. 7, no. 2, pp. 269–287, 2026. <https://doi.org/10.22437/mendapo.v7i2.55745>
- [50] Y. Setyawan, A. Erliyana, E. Makarim, F. A. Sjarif, L. R. Dewi, and A. N. A. Sukma, “Digital government post-reform in Indonesia: Normative developments and implementation by state organizing institutions,” *Law Reform*, vol. 21, no. 1, pp. 155–179, 2025. <https://doi.org/10.14710/lr.v21i1.68556>
- [51] H. Garg, “Some picture fuzzy aggregation operators and their applications to multicriteria decision-making,” *Arab. J. Sci. Eng.*, vol. 42, no. 12, pp. 5275–5290, 2017. <https://doi.org/10.1007/s13369-017-2625-9>
- [52] M. Batty, “Digital twins in city planning,” *Nat. Comput. Sci.*, vol. 4, no. 3, pp. 192–199, 2024. <https://doi.org/10.1038/s43588-024-00606-7>
- [53] Zainudin, “Digital ID policy in support of digital transformation in Indonesia,” *J. Bina Praja*, vol. 17, no. 1, pp. 1–18, 2025. <https://doi.org/10.21787/jbp.17.2025.1-18>
- [54] I. S. Ponto, J. Waisapy, and Z. Fadli, “Public service delivery in Indonesia’s outer islands: A performance evaluation,” *JPAP J. Penelit. Adm. Publ.*, vol. 11, no. 2, pp. 211–231, 2025. <https://doi.org/10.30996/jpap.v11i02.132484>
- [55] M. R. A. Farabi and A. Sintawati, “Flood early warning system at Jakarta dam using internet of things (IoT)-based real-time fishbone method to support industrial revolution 4.0,” *J. Soft Comput. Explor.*, vol. 5, no. 2, pp. 99–106, 2024. <https://doi.org/10.52465/josce.v5i2.293>
- [56] C. V. Marian, D. A. Mitrea, D. S. Rusu, and A. Vasilateanu, “Transparent digital governance: A blockchain-based workflow audit application,” *Appl. Sci.*, vol. 15, no. 21, p. 11694, 2025. <https://doi.org/10.3390/app152111694>
- [57] M. S. Ramadhan, M. Utama, I. Irsan, A. W. Anditya, and S. Aprita, “Algorithmic discrimination and the challenges of artificial intelligence regulation in Indonesia: A pluralistic justice perspective,” *J. Studi Sos. Polit.*, vol. 8, no. 2, pp. 158–173, 2024. <https://doi.org/10.19109/jssp.v8i2.36617>
- [58] D. Kurniawan, D. Mutiarin, A. Nurmandi, and E. P. Purnomo, “Interoperable organisations in crisis: Supporting data surveillance during Indonesia’s COVID-19 pandemic,” *Inf. Dev.*, 2025. <https://doi.org/10.1177/02666669251327350>
- [59] N. Notsu, S. Harada, and Y. Osaki, “Financial impacts of standardized system specifications: Evidence from Japanese municipalities,” *Int. Tax Public Financ.*, 2026. <https://doi.org/10.1007/s10797-026-09986-0>

Appendix

Table A1. First-place counts of the ten digital urban governance strategies across 10,000 Monte Carlo simulations for each weighting and ranking method combination

Ranking Method	Weighting Method	G_1	G_2	G_3	G_4	G_5	G_6	G_7	G_8	G_9	G_{10}	Dominant Strategy	Dominance (%)
TOPSIS	LOPCOW	0	0	0	89	0	0	9,911	0	0	0	G_7	99.11
TOPSIS	Entropy	0	0	0	3,961	0	0	110	2,917	0	3,012	G_4	39.61
TOPSIS	MEREC	0	0	0	640	0	0	243	1,572	0	7,545	G_{10}	75.45
TOPSIS	CRITIC	0	0	0	0	0	0	0	144	0	9,856	G_{10}	98.56
VIKOR	LOPCOW	53	0	0	0	0	0	9,947	0	0	0	G_7	99.47
VIKOR	Entropy	0	0	0	2	0	0	0	9,997	0	1	G_8	99.97
VIKOR	MEREC	0	0	0	0	0	0	0	10,000	0	0	G_8	100
VIKOR	CRITIC	0	0	0	0	0	0	0	9,920	0	80	G_8	99.2
MOORA	LOPCOW	0	0	0	2,685	0	0	7,315	0	0	0	G_7	73.15
MOORA	Entropy	0	0	0	7,418	0	0	301	0	0	2,281	G_4	74.18
MOORA	MEREC	0	0	0	3,704	0	0	442	0	0	5,854	G_{10}	58.54
MOORA	CRITIC	0	0	0	0	0	0	0	0	0	10,000	G_{10}	100
MABAC	LOPCOW	0	0	0	2,624	0	0	7,376	0	0	0	G_7	73.76
MABAC	Entropy	0	0	0	6,790	0	0	149	0	0	3,061	G_4	67.9
MABAC	MEREC	0	0	0	2,332	0	0	162	0	0	7,506	G_{10}	75.06
MABAC	CRITIC	0	0	0	0	0	0	0	0	0	10,000	G_{10}	100

(continued)

(continued)

WASPAS	LOPCOW	0	0	0	7	0	0	9,993	0	0	0	G_7	99.93
WASPAS	Entropy	0	0	0	332	0	0	708	8,960	0	0	G_8	89.6
WASPAS	MEREC	0	0	0	0	0	0	755	9,245	0	0	G_8	92.45
WASPAS	CRITIC	0	0	0	0	0	0	0	10,000	0	0	G_8	100
EDAS	LOPCOW	0	0	0	5,496	0	0	4,504	0	0	0	G_4	54.96
EDAS	Entropy	0	0	0	6,524	0	0	81	0	0	3,395	G_4	65.24
EDAS	MEREC	0	0	0	3,026	0	0	127	0	0	6,847	G_{10}	68.47
EDAS	CRITIC	0	0	0	0	0	0	0	0	0	10,000	G_{10}	100
ARAS	LOPCOW	0	0	0	3,351	0	0	6,649	0	0	0	G_7	66.49
ARAS	Entropy	0	0	0	6,532	0	0	162	0	0	3,306	G_4	65.32
ARAS	MEREC	0	0	0	3,002	0	0	208	0	0	6,790	G_{10}	67.9
ARAS	CRITIC	0	0	0	0	0	0	0	0	0	10,000	G_{10}	100
MARCOS	Hybrid	0	0	0	2,725	0	0	7,275	0	0	0	G_7	72.75

Note: TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; VIKOR = Multicriteria Optimization and Compromise Solution; MOORA = Multi-Objective Optimization by Ratio Analysis; MABAC = Multi-Attributive Border Approximation Area Comparison; WASPAS = Weighted Aggregated Sum Product Assessment; EDAS = Evaluation based on Distance from Average Solution; ARAS = Additive Ratio Assessment; MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; LOPCOW = Logarithmic Percentage Change-driven Objective Weighting; MEREC = Method based on the Removal Effects of Criteria; CRITIC = Criteria Importance Through Intercriteria Correlation; RANCOM = Ranking Comparison; Hybrid = combined LOPCOW–RANCOM weighting with $\alpha = 0.50$; G_i = the i th digital governance strategy ($i = 1, \dots, 10$); Dominance (%) = percentage of simulations in which the dominant strategy ranked first.